

Reinforcer Postposing Revised:
Linearization and Exponence at the Syntax-Phonology Interface

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Abstract: This paper investigates the nature of linearization and exponence by documenting and analyzing an output-optimizing alternation between long-distance displacement, suppression, and regular phonological repairs. Mandar (Austronesian, Central Indonesia) has a pair of monosyllabic locative adverbs that are routinely generated in DPs, forced to shift to the edge of the utterance, and suppressed when they cannot move. These steps are phonologically transparent in three respects: they ensure that these adverbs are only exponed at right edge of a phonological domain, they visibly resolve an output constraint, and they are suspended when the prosodic context enables a more optimal phonological repair. The paper thus argues that the systems of linearization and exponence must be split into two parts: a morphological stage that linearizes the terminal nodes of the syntax and matches them with vocabulary entries in a manner roughly familiar from Distributed Morphology (Halle & Marantz, 1993) and a phonological stage that reinstantiates this linear order and inserts the exponents that the morphology selects in the parallel, global, and output-optimizing calculus where prosodic structure is built (Selkirk, 2009, 2011).

Keywords:

Linearization, Exponence, Distributed Morphology, Match Theory, Prosodic Hierarchy

1 Introduction

It is a central target of minimalist inquiry to understand the structure of the PF branch: the series of feed-forward modules that translate syntactic representations into phonetic strings. In the domain of constituency, many lines of work have converged to suggest that representations undergo a sequential process of imperfect translation as they pass through the modules of this branch. The literature on Distributed Morphology (Halle & Marantz, 1993) for instance, has established that the terminal nodes of the syntax must be parsed into morphological words and subwords (Embick & Noyer, 2001) and then reworked through operations like node insertion, fission, fusion, and obliteration in an ordered and sequential morphology (Embick, 2000; Embick & Noyer, 2001; Arregi & Nevins, 2012; Hewett, 2023). The literature on the Prosodic Hierarchy, in turn, has established a similar network of results: morphological words and morphosyntactic phrases must be reinstantiated as prosodic words and phonological phrases in a parallel and global phonological calculus that reworks the ensuing phonological constituency in response to output constraints (Nespor & Vogel, 1986; McCarthy & Prince, 1993).

The goal of this paper is to advance our understanding of two systems that play out along this branch. The first is the system of Linearization, which establishes relations of linear precedence over the output correspondents of the unordered terminals of the syntax (Chomsky, 1995). The second is the system of Exponence, which associates these nodes with phonological content—in the terminology of Distributed Morphology, through a process of vocabulary insertion that pairs them with vocabulary entries from the lexicon (Halle & Marantz, 1993). The literature has established that certain parts of these systems must operate in the morphology: many patterns of allomorph selection are sensitive to information that is only available in that module (Halle & Marantz, 1993), many steps of movement seem to proceed over morphological representations

(Embick & Noyer, 2001; Embick, 2007; Arregi & Nevins, 2012; Myler, 2013), and many movements of this type feed allomorph selection (Adger, 2006; Ostrove, 2018). But there are reasons to believe that the phonology must play a role in each system as well: there are steps of movement that seem sensitive to prosodic constituency (Halpern, 1995; Bennett *et al.*, 2016; Aissen, 2017; Kusmer, 2019), patterns of exponence that reference global phonological context (McCarthy, 2011; Henderson, 2012; Brodtkin, 2025b), and patterns of both types that seem to repair output constraints (Mester, 1994; Kager, 1996; Perlmutter, 1998; Booij, 1998; Rubach & Booij, 2001; Mascaró, 1996b, 2007; Kurisu, 2001).

To refine the ensuing theoretical picture, this paper aims to document and analyze a system that shows optimizing alternations between displacement, non-exponence, and a regular phonological repair. This is one that governs the distribution of two locative adverbs in Mandar, an Austronesian language of Sulawesi. These adverbs are lexically selected by a subset of demonstratives and surface after many DPs at the edges of utterance-sized domains, along the lines in (1). In the terminology of Bernstein 1997, I will refer to these adverbs as REINFORCERS and their associated DPs as ASSOCIATES.

(1) *The Reinforcers*

- a. De tomauweng *(e)
 this grey one here
 ‘This old dude *(here)’
- b. Ka’bal i de’ [_{DP} **do** tomauweng **o**].
 invincible 3ABS they say that grey one there
 ‘That old dude there is invincible, they say.’ Sikki *et al.* 1987, 541

The analytical puzzle emerges from three interlocking effects. When their associates surface in utterance-medial positions, first, the reinforcers are typically forced to shift to the right (2a). When the right edge is occupied by other reinforcers, second, they are typically not exponed (2b). Under specific phonological circumstances, third, both

processes are suspended and the reinforcers begin to surface in utterance-medial positions after their associated DPs (2c). I will refer to these patterns of movement and suspended exponence as Reinforcer Postposing and Reinforcer Deletion.

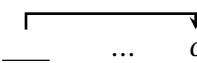
(2) *Reinforcer Postposing and Reinforcer Deletion*

- a. Naita i [DP **do** guru ____] [DP iting sanaeke] **o**.
 saw that teacher there that kid
 ‘That teacher there saw that kid.’
- b. Naita i [DP **do** guru ____] [DP **de** sanaeke e].
 saw that teacher there this kid here
 ‘That teacher there saw this kid here.’
- c. Mane naita i [DP **do** guru **o**] [DP **de** sanaeke e].
 just saw that teacher there this kid here
 ‘That teacher there just saw this kid here.’

The first task of the paper is to refine and extend the initial case, laid out in [Brodkin 2026](#), that Reinforcer Postposing occurs in the phonology. I lay out three arguments for this analysis: it ignores the locality constraints of the syntax, it proceeds along a path that can only be defined in phonological terms, and it is rerouted by changes in prosodic constituency that can only be forced by output constraints. The combined force of these arguments is to suggest that it carries its targets to the edge of the largest domain of the Prosodic Hierarchy: the intonational phrase ([Nespor & Vogel, 1986](#); [Selkirk, 2009](#)).

(3) *Phonological Displacement*

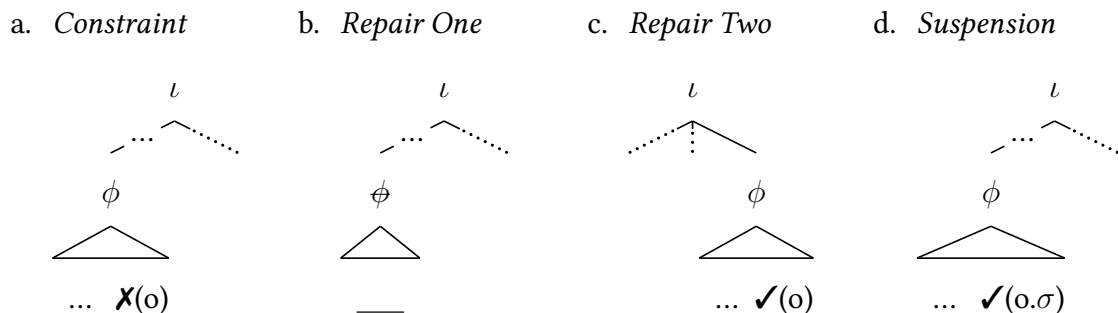
{_u ... do NP ____ ... o }



The second task is to establish that Reinforcer Postposing and Reinforcer Deletion operate as item-specific repairs to a general output constraint. The case proceeds along the same logical path. First, I show that Reinforcer Postposing and Deletion are phonologically transparent: the reinforcers violate a constraint against phrase-final

monosyllabic feet (4a) and the phonology resolves it by suspending exponence (4b) or moving them to positions that license these feet (4c). Second, I show that both processes are suspended when the driving constraint can be resolved in a different way (4d).

(4) *Reinforcer Deletion and Postposing: Output Optimization*



To capture these effects, I propose that linearization and exponence must each be split into two parts. In the morphology, I propose that the terminal nodes of the syntax are subjected to a first round of linearization and matched with vocabulary entries from the lexicon. In the phonology, I then propose that vocabulary entries are inserted into these terminal nodes and the linearization of the morphology is reimposed over their output correspondents in a characteristically imperfect way (and see also [Bye & Svenonius 2012](#); [De Belder 2020](#); [Bennett et al. 2016](#); [Kusmer 2019](#)). The result is a framework that allows us to understand patterns of phonological relinearization and phonologically conditioned exponence as cases of morphosyntax-phonology mismatch.

The remainder of this paper is organized as follows. Section 2 provides background information on Mandar and the reinforcers. Section 3 introduces Reinforcer Postposing and Reinforcer Deletion, partially reviewing the results of [Brodkin 2026](#). Section 4 shows that both processes provide item-specific repairs to an output constraint and Section 5 develops a formal analysis. Section 6 then concludes with a cross-modular link to the theory of Greed ([Chomsky, 1995](#); [Bošković, 1995](#)).

2 Background

Mandar is spoken by roughly 400,000 people in the Indonesian province of West Sulawesi. It is a language of the South Sulawesi subgroup, a primary subgroup of Malayo-Polynesian (Esser, 1938; Mills, 1975), and its closest relatives have been described in a small body of English-language work (Friberg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Lee, 2008; Kaufman, 2008; Laskowske, 2016; Finer, 1997, 1998, 1999; Béjar, 1999). This paper studies the standard variety of the language, which is spoken along the southern coastal plain of West Sulawesi (Grimes & Grimes, 1987) and which has been described in a grammar (Pelenkahu *et al.*, 1983) and discussed in many further publications by the Indonesian language office of South Sulawesi (Sikki *et al.*, 1987; Muthalib & Sangi, 1991; Friberg & Jerniati, 2000).

The data in this paper has been gathered through elicitation with a single speaker from the town of Ugibaru, Jupri Talib, with whom I have worked continually since 2018. The patterns of movement and deletion were encountered in 2019, documented in 2021, cross-checked in the field in 2022, and laid out in Brodtkin 2026. The phonological systems that drive and suspend these effects, which are the focus of this paper, have come to light much more slowly through this time. Like the patterns of movement and deletion, however, they seem easy to replicate in many other dialects of Mandar and in several languages nearby, like Pattae'. I leave regional discussion to future work.

2.1 Clausal Syntax

Mandar is a head-marking and head-initial language that shows a consistent basic word order of vso. It shows an ergative-absolutive system of agreement: transitive verbs agree with the s and second-position clitics agree with the transitive o and intransitive s.

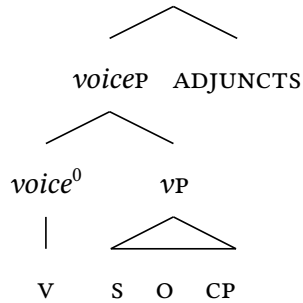
The language requires postverbal arguments to surface before complement clauses and requires complement clauses to precede locative and temporal adjuncts to the *voiceP* (5).¹

(5) *The Mandar Clause*

- a. U-irrangngi i [s yau] [CP mua' lamba i marondong] dionging.
 1ERG-hear 3ABS 1SG that go 3ABS tomorrow yesterday
 'I heard yesterday that he was going tomorrow.'
- b. U-pissang i [s yau] [o iKaco'] [CP mua' pole a'] marondong.
 1ERG-tell 3ABS 1SG NAME that come 1ABS tomorrow
 'I'll tell Kacho' that I'll come tomorrow.'

Brodkin 2025a shows that the vso string forms a tight domain that corresponds to the extended VP. In this space, the verb consistently raises to a high head at the edge—by hypothesis, *voice*⁰—and all following arguments and complement clauses are realized in their base positions in the *voiceP*. The locative and temporal adjuncts that follow the vso-CP string are adjoined to the *voiceP*, yielding the rough clausal syntax in (6).

(6) *Mandar Clausal Syntax*



2.2 The Demonstrative-Reinforcer Construction

Mandar has four regular locative adverbs: *indi* 'right here', *dini* 'here', *dio* 'there', and *diting* 'over there'. It also has two monosyllabic locative adverbs that are synonymous

with *dini* and *dio*: *e* ‘here’ and *o* ‘there’. All of these elements can be independently adjoined to the *voiceP*, as in (7a)-(7b), and the adverbs of the two sets can co-occur (7c).

(7) *The Locative Adverbs*

- a. Urang i [**indi** / **dini** / **dio** / **diting**].
rain 3ABS right here here there over there
‘It’s raining [right here / here / there / over there].’
- b. Urang i [**e** / **o**].
rain 3ABS here there
‘It’s raining [here / there].’
- c. Mala bappa i [*voiceP* na-tarima akkatta-ta’] **diting** **o**.
can hopefully 3ABS 3ERG-accept purpose-2GEN over there there
‘Hopefully they can accept your purpose there.’ Friberg & Jerniati 2000, 243

The language also has four demonstratives that draw the same contrast in distance: *ndi* ‘this’ (very close), *de* ‘this’, *do* ‘that’, and *iting* ‘that’ (out of sight). Both types of locative adverbs can appear in the *DPS* that host these demonstratives to yield the demonstrative-reinforcer constructions below (Bernstein, 1997; Roehrs, 2010).

(8) *Demonstrative-Reinforcer Constructions*

- | | |
|---|--|
| a. Ndi posa indi e
this cat here here
‘This cat here’ | c. Do posa dio o
that cat there there
‘That cat there’ |
| b. De posa dini e
this cat here here
‘This cat here’ | d. Iting posa diting o
that cat there there
‘That cat there’ |

In fragments of this type, the adverbs *e* and *o* are forced to appear after every *DP* that contains one of the demonstratives *ndi*, *de*, or *do*. The adverb *e* is required after *ndi* and *de* and the adverb *o* is required after *do*. In each case, it is ungrammatical to drop the

adverb or replace it with anything else: *e* cannot be replaced with *o*, *o* cannot be replaced with *e*, and neither can be replaced with any of the regular locative adverbs.²

(9) *Selectional Requirements*

- | | |
|--|--|
| <p>a. De posa *(e)
 this cat here
 ‘This cat *(here)’</p> | <p>b. Do posa *(o)
 that cat there
 ‘That cat *(there)’</p> |
|--|--|

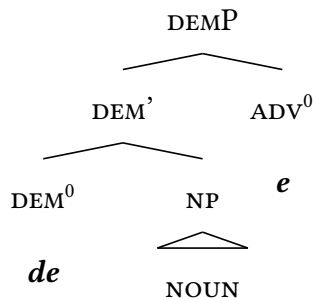
These requirements are specific to *ndi*, *de*, and *do*. Neither adverb is ever required beneath *iting* or any D⁰-like element in the language, such as the proprial article *i*.

(10) *Lexical Idiosyncrasy*

- | | |
|--|--|
| <p>a. [_{DP} Itiŋ posa (o)].
 that cat there
 ‘That cat (there).’</p> | <p>b. [_{DP} i Kaco’ (e)].
 D NAME here
 ‘Kacho’ (here).’</p> |
|--|--|

Brodkin 2026 thus proposes that *e* and *o* are LEXICALLY SELECTED by the demonstratives *ndi*, *de*, and *do*: these demonstratives spell out a head DEM⁰ that systematically introduce *e* and *o* in their specifiers. In this context, I will refer to *e* and *o* as REINFORCERS and the DPS that contain them as their ASSOCIATES (Bernstein, 1997).

(11) *The Syntax of Selection*



2.3 Separation

Outside of fragments, the reinforcers surface in just the same way when their associates fall at the right edges of matrix clauses (12a). When their associates take clause-medial

positions, however, we typically see the effect in (12b): outside of a configuration that depends on the distribution of prosodic words across the clause and the metrical status of the syllable that follows the associate (Section 4), they shift on their own to the right. I will refer to this process as REINFORCER POSTPOSING below.

(12) *Movement to the Right*

- a. Na-saka i [DP do posa **o**].
 3ERG-catch 3ABS that cat there
 ‘That cat there caught it.’
- b. Na-saka i [DP do posa ____] balao **o**.
 3ERG-catch 3ABS that cat mouse there
 ‘That cat there caught the mouse.’

Reinforcer Postposing invariably places the reinforcers at the right edge of a clause-like domain. To this end, it routinely draws them across arguments (13a), adjuncts (13b), coordinate structures (13c), and complement CPS (13d).

(13) *Crossing Arguments and Adjuncts*

- a. Mane na-pande i [DP do Kaco’ ____] bau posa **o**.
 just 3ERG-feed 3ABS that NAME fish cat there
 ‘That Kacho’ there just fed the cat a fish.’
- b. Mam-buno i [DP de posa ____] ulo kiang dio di bongi **e**.
 ANTIP-kill 3ABS that cat snake big there at night here
 ‘This cat here killed a big snake there last night.’
- c. Na-peroa i [DP do tau ____] [CONJP iKaco’ na iCicci’] **o**.
 3ERG-invite 3ABS that person NAME and NAME there
 ‘That person there invited Kacho and Chichi.’
- d. Ma’ua i [DP do Kaco’ ____] [CP mua’ macoa i mua’ lamba a’] **o**.
 said that NAME that good if go 1ABS there
 ‘That Kacho’ there said that it would be good if I went.’

3 Phonological Displacement

The simplest analysis of this effect would treat it as a type of syntactic movement to the right, on a par with the RIGHTWARD FOCUS MOVEMENT in (14): one that targets certain types of foci and places them in positions linearized to the right of the *voiceP*.

(14) *Rightward Focus Movement*

- a. $[_{FOCP} [_{voiceP} \text{U-baca} \quad i \quad ___] \text{dionging} [_o \text{BUKU-NNA} \text{ iCICCI'}]]$.
 1ERG-read 3ABS yesterday book-3GEN NAME
 ‘I read CHICHI’S BOOK yesterday.’
- b. $[_{FOCP} [_{voiceP} \text{U-baca} \quad i \quad [_{DP} [_D' \text{buku-nna}] ___]] \text{dionging} \text{ iCICCI'}]$.
 1ERG-read 3ABS book-3GEN yesterday NAME
 ‘I read CHICHI’S book yesterday.’

This section shows that Reinforcer Postposing must instead occur in the phonology, drawing the reinforcers across phonological representations to positions defined in surface prosodic terms. The case proceeds in three parts (and see [Brodkin 2026](#) for partially overlapping arguments to this end). First, the reinforcers seem to move at a stage that lacks syntactic information, allowing them to escape islands and ignore c-command. Second, the reinforcers seem to move at a stage that has access to prosodic structure—suggesting that they must move after the morphosyntax. Third and finally, they seem to react to phonologically induced changes in prosodic constituency, suggesting that they shift in the phonological calculus where prosodic structure is built.

3.1 Island Effects

The work begins in the arena of absolute locality. Mandar shows a standard range of island constraints that prohibit extraction from many syntactic domains: for instance,

from coordinate structures (15a) and complex NPs (15b). Rightward Focus Movement cannot escape from any islands of this type.

(15) *Syntactic Islands*

- a. U-ita i [_{CP} sola-na iKaco' na sola-na iCicci'] allo ajuma'.
 1ERG-see 3ABS friend-3GEN NAME and friend-3GEN NAME friday
 'I saw Kacho's friend and Chichi's friend on Friday.'
- b. Ma'ita a' [_{NP} karewa [_{CP} mua' pole i sola-na iCicci']] dio di FB.
 I saw news that come 3ABS friend-3GEN NAME there on FB
 'I saw [_{NP} news [_{CP} that Chichi's friend is coming]] on facebook.'

The reinforcers can escape all of these domains: for instance, the conjuncts of coordinate structures (16a)-(16b) and clauses embedded in NPs (16c).

(16) *Reinforcer Postposing: Escapes Islands*

- a. U-ita i [_{CP} sola-na iKaco' na do tau ____] allo ajuma' o.
 1ERG-see 3ABS friend-3GEN NAME and that guy friday there
 'I saw Kacho's friend and that guy there on Friday.'
- b. U-ita i [_{CP} do tau ____ na sola-na iKaco'] allo ajuma' o.
 1ERG-see 3ABS that guy and friend-3GEN NAME friday there
 'I saw that guy there and Kacho's friend on Friday.'
- c. Ma'ita a' [_{DP} karewa [_{CP} mua' pole i do tau ____]] dio di FB o.
 I saw news that come 3ABS that guy there on FB there
 'I saw [_{DP} news [_{CP} that that guy there is coming]] on facebook.'

They also evade the Right Roof Constraint, which prevents rightward movement from embedded CPs from crossing elements in the matrix clause (Ross 1967). When embedded clauses are followed by right-adjuncts to the matrix *voiceP*, then, they form islands for Rightward Focus Movement (17a) but not for Reinforcer Postposing (17b).

(17) *Reinforcer Postposing: Crosses Right Rooves*

- a. *Pepeissang i [_{CP} mua' pole i sola-na ____] **dio di facebook** iCicci'.
 Go check if come 3ABS friend-3GEN there on facebook NAME
 INTENDED: 'Go check [_{CP} if CHICHI's friend is coming] on facebook.'

- b. Pepeissang i [CP mua' pole i do tau ____] **dio di facebook o.**
 Go check if come 3ABS that guy there on facebook there
 'Go check [CP if that guy there is coming] on facebook.'

3.2 Relative Locality

These effects are matched by a parallel deviation in the domain of Relative Locality.

When multiple reinforcers are generated in a single domain, they surface after their associates when they can remain in situ. As a result, there are configurations where reinforcers surface after multiple arguments in the *voiceP* (18a) or multiple constituents in the DP, like the head noun and a following possessor in SPEC,DP (18b).

(18) *Single Domains: Multiple Reinforcers*

- a. Mane na-bengang i [DP de guru e] buku-na [DP do to basa o].
 just 3ERG-give 3ABS this teacher here book-3GEN that linguist there
 'This teacher here just gave his book to that linguist.'
- b. [DP De buku-na e [DP guru-na [DP do sanaeke maroca' sannal o]]]
 this book-3GEN here teacher-3GEN that kid loud very
 'This book here of the teacher of those really loud kids.'

When multiple reinforcers are generated in positions that force movement, however, we see the following pair of effects. First, Reinforcer Postposing applies to the reinforcers that would otherwise surface linearly farthest to the right: those generated in the most deeply embedded rightward specifiers in the DP (19a) and the rightmost arguments in the *voiceP* (19b). Second, all other reinforcers are forced to delete.

(19) *Reinforcer Competition*

- a. Na-bengang i [DP de guru ____] buku-na [DP do to basa ____] digena' o.
 3ERG-give 3ABS this teacher book-3GEN that linguist earlier there
 'This teacher (here) gave his book to those loud kids earlier.'
- b. [DP De buku-na ____ [DP guru-na [DP do ana' ____]]] dio di meja] o.
 this book-3GEN teacher-3GEN that kid there on table there
 'This book (here) of the teacher of those kids there that's on the table.'

The relative locality constraints that guide this step are sensitive to linear order alone. As a result, they routinely forego reinforcers in structurally higher positions for reinforcers generated in lower positions that should surface farther to the right. The following examples illustrate this effect in VS-PP-X strings where reinforcers are generated in both subjects and PPs: when the reinforcers cannot remain in situ, the reinforcers in the PPs shift and the reinforcers in the subjects are not exponed.

(20) *Reinforcer Competition: Suppression*

- a. Sidego i [DP do putra ____] sola [DP ndi putri ____] di wongi e.
 dance 3ABS that prince with this princess at night here
 ‘That prince (there) danced with this princess here last night.’
- b. Di-anna i [DP de kocci’ ____] tama [DP do kota’ ____] digena’ o.
 PASS-put 3ABS this key into that box earlier there
 ‘This key (here) was placed in that box there earlier.’

3.3 Phonological Transparency

These facts converge with two further effects to suggest that Reinforcer Postposing and Reinforcer Deletion operate over the hierarchical structure of the phonology (Selkirk 1984, 1986; Nespor & Vogel 1986). Following Itô & Mester 2007, 2009, 2012, 2013, I take this constituent structure to be built from the narrow inventory of abstract and universal prosodic domains in (21): the metrical mora, syllable, and foot (Lieberman & Prince, 1977; Hayes, 1981), the interface categories of the prosodic word and phonological phrase (McCarthy & Prince, 1993; Truckenbrodt, 1995, 1999, 2002), and the speech-act category of the intonational phrase (Myrberg, 2010, 2013; Hamlaoui & Szendrői, 2015, 2017).

(21) *The Prosodic Hierarchy*

μ	>	σ	>	FT	>	ω	>	ϕ	>	ι
MORA		SYLLABLE		FOOT		WORD		PHRASE		INTONATIONAL PHRASE

The first argument begins from the diagnostics for the Intonational Phrase. Mandar has a process of Nasal Assimilation that assimilates coda /ŋ/ to all following segments in place and then denasalizes it before all but /m n p ŋ/ and /b d d̥g/ (Mills, 1975; Pater, 1999). The following examples show how this process works in the prosodic word, using the top line to show its surface effect and the second line to show underlying forms. In affixes that end in /ŋ/, this segment always assimilates in place to following /b d d̥g/ (22a) and then denasalizes before /p t t̥k/ (22b).³

(22) *Nasal Assimilation*

- | | | | | |
|----|--|-----------------|--------------------|------------------|
| a. | mambawa, | mindai? | taŋd̥ʒappo? | saŋgallas |
| | maŋ-bawa | miŋ-dai? | taŋ-d̥ʒappo? | saŋ-gallas |
| | ANTIP-bring | VERBALIZER-up | NEG-destroy | ONE-glass |
| | ‘bring, ascend, undestroyed, one glass.’ | | | |
| | | | | |
| b. | mappolon, | mittama, | tatt̥ʃappu? | sakkat̥ʃa |
| | maŋ-polon | miŋ-tama | taŋ-t̥ʃappu? | saŋ-kat̥ʃa |
| | ANTIP-cut | VERBALIZER-in | NEG-exhausted | ONE-cup |
| | ‘cut, enter, unexhausted, one cup.’ | | | |

This process applies between words in all the strings that are not split by any obvious pause: for instance, fragments (23a) and root clauses that contain strings of arguments (23b), coordinate structures (23c), and embedded cps (23d) within the *voiceP*.

(23) *Nasal Assimilation: Through the Utterance*

- | | | | | | |
|----|---|---------------------------|---------------------------|------------------------|----------------------|
| a. | { _u te.ɪos | sola sapik | kes | sannal } | |
| | tedoŋ | sola sapin | kaiaŋ | sanna ¹ | |
| | water | buffalo | and cow | big very | |
| | ‘Very big water buffalo and cows.’ | | | | |
| | | | | | |
| b. | { _u pure | napolop | pa? ramar | roppot | te?e } |
| | pura i | na-poloŋ | pua? ramaŋ | roppoŋ | te?e |
| | done 3ABS | [_v 3ERG-cut] | [_s MR. NAME] | [_o grass] | [_x now] |
| | ‘Mr. Ramang is finished cutting the grass now.’ | | | | |

- c. {_ℓ malli iramat te.ɾos sola sapit ton djolo? }.
 maŋ-alli i iramaŋ tedoŋ sola sapin taun diolo?
 ANTIP-buy 3ABS [_s NAME] [_{CONJP} buffalo and cow] [_x year last]
 ‘Ramang bought buffalos and cows last year.’
- d. {_ℓ wirraŋŋi ma? pole iramam marondot ton djolo? }.
 u-irraŋŋi i mua? pole i iramaŋ marondoŋ taun diolo?
 1ERG-hear 3ABS [_{CP} that come 3ABS NAME tomorrow] year last
 ‘I heard that Ramang was coming tomorrow a year ago.’

Despite this fact, it is blocked at a series of junctures that host a type of pause that I will term Comma Intonation: one that has the same phonetic shape and surface distribution as the prosodic juncture marked as ToBi Break Index 4 in English. This type of pause appears systematically around four types of syntactic domains. The first are left-peripheral topics, which are always followed by Comma Intonation whether they are raised to the left periphery (24a) or base-generated in that domain and resumed with pronouns below (24b). At their right edges, in each case, Nasal Assimilation is blocked.

(24) *Left-Peripheral Topics*

- a. {_ℓ iramaŋ }, {_ℓ ta?wita i }.
 iramaŋ ta?-u-ita i
 NAME NOT-1ERG-see 3ABS *t*
 ‘Ramang, I didn’t see.’
- b. {_ℓ me ramaŋ }, {_ℓ ta?wita i }.
 mua? iramaŋ ta?-u-ita i
 as for NAME NOT-1ERG-see 3ABS *pro*
 ‘As for Ramang, I didn’t see him.’

The second are the right-peripheral topics in (25), which are base-generated in high positions, linearized after the *voiceP*, and resumed by pronouns (at least when nominal) in the preceding clause. These topics form islands for Nasal Assimilation as well.

(25) *Right-Peripheral Topics*

- a. {_ℓ nete iramaŋ }, {_ℓ kindo?na? }
 na-ita i iramaŋ kindo?-na
 3ERG-see 3ABS NAME *pro* mom-3GEN
 ‘Ramang saw her, his mom.’
- b. {_ℓ nete iramaŋ }, {_ℓ ton djolo? }
 na-ita i iramaŋ tauŋ diolo?
 3ERG-see 3ABS NAME year last
 ‘Ramang saw her, last year.’

The third are appositives and parentheticals (26a) and the fourth are initial adjunct CPs, which behave like their counterparts in English: they can either be set off by Comma Intonation or not (26b)-(26c). When these constituents are offset, they are opaque to Nasal Assimilation; when initial adjunct CPs are not, they are transparent instead.

(26) *Appositives and Initial Adjunct CPs: Parallel Effects*

- a. {_ℓ nete iramaŋ }, {_ℓ kapala ɾesa }, {_ℓ ton djolo? }.
 na-ita i ramaŋ kapala desa tauŋ diolo?
 3ERG-see 3ABS NAME head village year last
 ‘She saw Ramang, the mayor, last year ’
- b. {_ℓ ma? pole iramaŋ }, {_ℓ pole toa? }.
 mua? pole i iramaŋ pole to a?
 if come 3ABS NAME come also 1ABS
 ‘If Ramang’s coming, I’m coming.’
- c. {_ℓ ma? pole iramaŋ pole toa? jau }.
 mua? pole i iramaŋ pole to a? iau
 if come 3ABS NAME come also 1ABS 1SG
 ‘If Ramang’s coming I’M ALSO coming.’

These facts suggest that Nasal Assimilation and Comma Intonation are keyed to the right edge of an abstract prosodic domain that is built around complete utterances (27a)-(27b) and a narrow set of syntactic domains: left- and right-peripheral topics

(27c)-(27e), appositives and parentheticals (27f), and most initial adjunct cps (27g)-(27h).

This is the Intonational Phrase.⁴

(27) *The Intonational Phrase*

- | | |
|--|--|
| a. { _ι My sister }. | e. { _ι { _ι I'd see her }, { _ι my sister } }. |
| b. { _ι I saw my sister }. | f. { _ι { _ι I'd see Zoe }, { _ι my sister } }. |
| c. { _ι { _ι My sister }, { _ι I saw } }. | g. { _ι { _ι If my sister went }, { _ι I'd go } }. |
| d. { _ι { _ι As for my sister }, { _ι I saw her } }. | h. { _ι If my sister went I'd GO }. |

From this position, we can establish three layers of phonological structure around the process of Reinforcer Postposing. First, the reinforcers remain in-situ when their associates are final in the intonational phrase. As a result, they follow the associates that are left-peripheral topics (28a)-(28b), right-peripheral topics (28c), and appositives (28d).

(28) *Reinforcers: Follow Associates that are Final in the Intonational Phrase*

- a. { do ramang o? }, { wita i }.
do ramang o u-ita i
that NAME there 1ERG-see 3ABS t
‘That Ramang there, I saw.’
- b. { ma? do ramang o? }, { wita i }.
mua? do ramang o u-ita i
as for that NAME there 1ERG-see 3ABS pro
‘As for that Ramang there, I didn’t see him.’
- c. {_ι nete solau ton djolo? }, {_ι do ramang o? }.
na-ita i sola-u taung diolo? do ramang o
3ERG-see 3ABS pro friend-1GEN year last that NAME there
‘He saw my friend last year, that Ramang there.’
- d. {_ι nete kapala desa }, {_ι do ramang o? }, {_ι ton djolo? }.
na-ita i kapala desa do ramang o taung diolo?
3ERG-see 3ABS head village that NAME there year last
‘She saw the mayor, that Ramang there, last year ’

Second, Reinforcer Postposing always proceeds from positions that are medial in the intonational phrase and lands at its edge. As a result, it can place its targets at the edges of the strings that precede appositives (29a) and right-peripheral topics (29b) and the right edges of the adjunct cps that form their own intonational phrases ((29c); cf. (29d)).

(29) *Reinforcers: Move to the Right Edge of the Intonational Phrase*

- a. {_l nete io ramas _____ solau **oʔ** }, {_l ton djoloʔ }.
 na-ita i do ramaŋ sola-u o tauŋ dioloʔ
 3ERG-see 3ABS that NAME friend-1GEN there year last
 ‘That Ramang there saw my friend, last year.’
- b. {_l nete io tau _____ iramaŋ **oʔ** }, {_l kapala ɿesa }, {_l ton djoloʔ }.
 na-ita i do tau iramaŋ o kapala desa tauŋ diolo
 3ERG-see 3ABS that person NAME there head village year last
 ‘That person there saw Ramang, the mayor, last year’
- c. { maʔ pole io ramam _____ marondoŋ **oʔ** }, {_l pole toaʔ }.
 muaʔ pole i do ramaŋ marondoŋ o pole to aʔ
 if come 3ABS that NAME tomorrow there come also 1ABS
 ‘If that Ramang there is coming tomorrow, I’m coming too.’
- d. { maʔ pole io ramam _____ marondop pole toaʔ **oʔ** }.
 muaʔ pole i do ramaŋ marondoŋ pole to aʔ o
 if come 3ABS that NAME tomorrow come also 1ABS 1SG there
 ‘If that Ramang there’s coming tomorrow I’m ALSO coming.’

Third, Reinforcer Postposing always proceeds within the minimal intonational phrase. As a result, it cannot draw the reinforcers out of left-peripheral topics (30a)-(30b), appositives (30c), or prosodically independent initial adjunct cps (30d).

(30) *Reinforcers: Trapped within the Intonational Phrase*

- a. { do ramaŋ **oʔ** }, { wita i _____ }.
 do ramaŋ o u-ita i
 that NAME there 1ERG-see 3ABS t
 ‘That Ramang there, I saw.’

- b. { ma? do raman **o?** }, { wita i ____ }.
 mua? do raman o u-ita i
 as for that NAME there 1ERG-see 3ABS *pro*
 ‘As for that Ramang there, I saw him.’
- c. { nete kapala desa }, { do raman **o?** }, { ton djolo? ____ }.
 na-ita i kapala desa do raman o taun diolo
 3ERG-see 3ABS head village that NAME there year last
 ‘She saw the mayor, that Ramang there, last year’
- d. { ma? pole io raman **o?** }, { pole toa? ____ }.
 mua? pole i do raman o pole to a?
 if come 3ABS that NAME there come also 1ABS
 ‘If that Ramang there is coming, I’m coming.’

3.4 Phonological Rerouting

To this picture we can add one final fact. Prosodic constituency is routinely reworked in service of output constraints, and at the level of the intonational phrase, we can see this effect in the system of poetry. Mandar has a type of poem called a *Kalinda’da’* that is built around four metrical lines, along the lines in (31).

(31) *The Kalinda’da’*

- a. { ma? malo tolaen } { pammile mala?bi?na? }.
 mua? man-ala o tolaen pammilei i mala?bi?-na
 if ANTIP-take 2ABS someone else choose 3ABS better-3GEN
 ‘If you’d rather someone else, at least choose someone better.’
- b. { tane tolaen } { ma? sirrapanju di }.
 tania i tolaen mua? sin-rapan-u di
 is not 3ABS someone else if as-equal-1GEN just.3ABS
 ‘It’s not someone else if they’re just as bad as me.’ Muthalib & Sangi 1991: 12

To capture the facts of intonational phrasing in these poems, it may seem possible to assume that the metrical lines are separated by some process of the syntax: for instance,

the process of clausal extraposition in (33). This is one that operates beneath certain verbs in a lexically restricted way and forces its targets to form intonational phrases.

(32) *Clausal Extraposition*

- a. {_ℓ mo io ramam ____ ma? tallan i manini djonin **o?** }.
 man-ua i do raman mua? tallan i manini djonin o
 ANTIP-say 3ABS that NAME [CP that sink 3ABS later] yesterday there
 ‘That Ramang there said that it would sink later yesterday.’
- b. {_ℓ mo io raman ____ djonin **o?** } {_ℓ ma? tallan i manini }.
 man-ua i do raman djonin o mua? tallan i manini
 ANTIP-say 3ABS that NAME yesterday o [CP that sink 3ABS later]
 ‘That Ramang there said yesterday that it would sink later.’

In Mandar, however, it is clear that this analysis is incorrect. The case emerges from a freezing effect: when clauses are extraposed, they become islands for WH-movement. The following example illustrates this fact: WH-movement can escape from complement clauses that remain in the *voicer* (33a) but not the clauses that shift to the right (33b).

(33) *Extraposition: Freezing Effects*

- a. {_ℓ a no io ramat ____ tallam ____ manini djonin **o?** }?
 a na-ua do raman tallan manini djonin o
 what 3ERG-say that NAME [CP sink later] yesterday there
 ‘What did that Ramang there say would sink later yesterday?’
- b. {_ℓ *a no io raman ____ djonin **o?** } {_ℓ tallam ____ manini? }?
 a na-ua do raman djonin o tallan manini
 what 3ERG-say that NAME yesterday there [CP sink later]
 INTENDED: ‘What did that Ramang there say yesterday would sink later?’

The metrical lines that form their own intonational phrases do not show this effect. As a result, they are transparent to WH-movement: in the following poem, for instance, the WH-phrase *inna* ‘where’ moves from the second line into the first. I take this fact to suggest that the patterns of intonational phrasing in these poems are forced by purely phonological constraints that govern the number of syllables that appear in each line.

(34) *Wh-Movement: Crosses Metrical Lines*

- a. $\{_{\iota}$ **inna** wandamo .isaŋa } $\{_{\iota}$ diposara watammu? ____ }?
 inna bandamo di-saŋa di-posara batan-mu
 where exactly PASS-think PASS-spoiled body-2GEN
 ‘Where exactly do you think that you will be spoiled?’
- b. $\{_{\iota}$ atena wonji } $\{_{\iota}$ mala di pattindoan }.
 ate-na bonji mala di pattindoan
 heart-3GEN night can in bed
 ‘In the middle of night, it could happen in bed.’ Muthalib & Sangi 1991: 284

Reinforcer Postposing reacts to these constraints in a surface-transparent way: when the reinforcers are generated in strings that are parsed into multiple metrical lines, they shift to the edges of their associated lines instead of the positions that they would take without these splits. When the clause in (35a) is split by a linebreak, then, its reinforcer is repositioned in the poem recorded by Muthalib & Sangi 1991 in (35b).

(35) *Phonological Transparency*

- a. $\{_{\iota}$ jamo **io** ____ .isaŋa lopi pattonda roppon **o?** }
 jamo do ____ disaŋa lopi pattonda roppon **o** }
 is that called boat shipping grass there
 ‘That there is called a worthless endeavor.’
- b. $\{_{\iota}$ jamo **io** ____ .isaŋa **o?** } $\{_{\iota}$ lopi pattonda roppon }
 jamo do ____ disaŋa **o** } $\{_{\iota}$ lopi pattonda roppon }
 is that called there boat shipping grass
 ‘ $\{_{\iota}$ That there is called } $\{_{\iota}$ a worthless endeavor }.’ Muthalib & Sangi 1991: 374

We can leverage this effect to show that Reinforcer Postposing targets the right edge of the intonational phrase without regard to the syntactic shape of the landing site. By inserting reinforcers into other poems in Muthalib & Sangi 1991, for instance, we can force them to split strings of arguments in the *voiceP* (36a), complex DPS (36b), comparatives (36c), and the complements of consider- (36d) and perception-verbs (36e).

(36) *Phonological Rerouting*

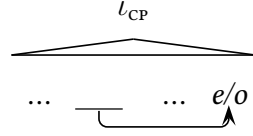
- a. {_ℓ sambo ɛe poppo? _____[✓] nawuŋ **e?** } {_ℓ passon di kappummu? }
 samboi i de pioppo? nauŋ e passauaŋ di kappuŋ-mu
 set 3ABS this cover down here well in village-2GEN
 ‘{_ℓ Set this cover here down } {_ℓ on the well in your village }.
- b. {_ℓ kaluppiŋa ã? **de** salla^mbar **e?** } {_ℓ garritin di lindomu? }
 kaluppiŋaŋ a? de saŋ-lambar e garritiŋ di lindo-mu
 wrap up for 1ABS this one-lock here hair from head-2GEN
 ‘{_ℓ Wrap up for me this lock here } {_ℓ of hair from your head }’
- c. {_ℓ moŋemoŋe pa? jau **e?** } {_ℓ na to nande ɣajaŋ }
 moŋe-moŋe? pa a? iau e na tau na-ande gaḍʒaŋ
 more woe to yet 1ABS me here than person 3ERG-eat sword
 ‘{_ℓ Even more woe to [_{DP} me here] } {_ℓ than one mauled by a sword }.
- d. {_ℓ naparrapam ma? jau **e?** } {_ℓ beluwelun di nawaŋ }
 na-parrapaŋ mo a? iau e belu-beluŋ di nabaŋ
 3ERG-consider already 1ABS me here firefly in sky
 ‘{_ℓ She already considers [_{DP} me here] } {_ℓ like a firefly in the sky }.
- e. {_ℓ wita wandi **de** saramu **e?** } {_ℓ sambar lo .i tau }
 u-ita bandi de sara-mu e sambar lao di tau
 1ERG-see really.3ABS this love-2GEN here slant toward in person
 ‘{_ℓ I see this love of yours here } {_ℓ falling away toward another }

4 Phonological Movement

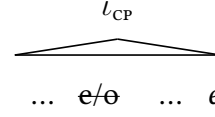
To capture these facts, I propose the analysis in (37): Reinforcer Postposing draws the reinforcers to the edge of the intonational phrase. I take this process to operate in the global, parallel, and constraint-based phonological calculus where prosodic structure is built, following the standard theory of prosodic structure-building (Selkirk 2009, 2011; for overviews, see also Inkelas & Zec 1995; Truckenbrodt 2007; Bennett & Elfner 2019). I propose that Reinforcer Deletion occurs at that stage as well: when the phonology cannot reposition the reinforcers in this way, it prevents them from being expounded.

(37) *The Phonological Analysis*

a. *Reinforcer Postposing*



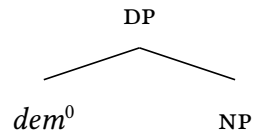
b. *Reinforcer Deletion*



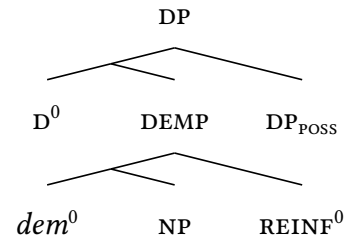
To integrate this result with the theory of linearization, I submit the following proposal. Following the standard view in Distributed Morphology, I assume that linear order is absent from the syntax (Chomsky, 1995) and established in the morphology (Embick, 2007; Kramer, 2010; Myler, 2013). Assuming the ordered morphological module of Arregi & Nevins 2012, I take this process to occur at an intermediate stage of the morphological derivation and establishes ordering relations in response to pressures of three types. The first are the pressures that determine head-complement order, which force heads to precede their complements in Mandarin (38a). The second are the pressures that linearize adjuncts and specifiers, which typically force them to fall to the left of their hosts (on which see López 2009b; Kusmer 2019, 2020). The third are the idiosyncratic preferences of particular heads, which have the language-internal effect in (38b): the specifiers that host possessors and reinforcers always surface on the right.

(38) *Morphological Linearization*

a.



b.

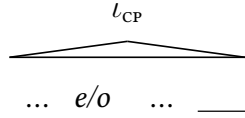


I propose that this initial linearization of terminal nodes is instantiated over their output constituents in the global and parallel phonological evaluation where prosodic structure is built. Following McCarthy & Prince 1993 and Halpern 1995, I take this

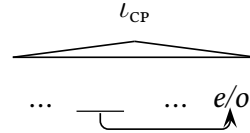
process of translation to be driven by ranked and violable constraints. This framework makes it possible for the phonology to directly rework the facts of linear order in response to output constraints: a position that allows us to capture the many patterns of phonological displacement that seem to operate in the prosodic word (Noyer, 1994; Hargus & Tuttle, 1997; Yu, 2007), across the phonological phrase (Halpern, 1992, 1995; Schütze, 1994a,b; King, 1996; Inkelas & Zec, 1990; Zec, 2005; Diesing & Zec, 2011, 2017; Kusmer, 2019), and in larger phonological domains (Bennett *et al.*, 2016; Aissen, 2017).⁵ In the system of Reinforcer Postposing, it also allows us to force the step in (39): one that relinearizes the reinforcers to the end of the intonational phrase whenever they are initially linearized into positions that would not fall at that edge.

(39) *Reinforcer Postposing: Phonological Relinearization*

a. *Input Linearization*



b. *Output Linearization*

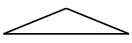
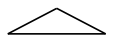


On this analysis, we can understand Reinforcer Postposing to reflect a mismatch between the linear order of terminal nodes in the morphology and the linear order of their exponents in the phonology. In the domain of constituency, syntax-prosody mismatches of this type are routinely forced by output constraints: in the terminology Optimality Theory (Prince & Smolensky, 2004), the restrictions on accentual clash (Kubozono, 1989, 1993; Selkirk & Elordieta, 2010), focus alignment (Féry, 2013, 2017; Féry & Ishihara, 2016), and balance and weight (Hayes & Lahiri, 1991; Ghini, 1993; Shih, 2016; Itô & Mester, 2019). The task of this section is to show that Reinforcer Postposing and Reinforcer Suppression are driven by two output constraints of this type: top-down requirements that are keyed to the right edges of large prosodic domains.

4.1 The Moraic Trochee

The work begins from the system of footing. At the level of the prosodic word, Mandarin shows a quantity-sensitive system of foot structure: when top-down constraints do not interfere, the language builds moraic trochees iteratively from the right edge of each prosodic word. These types of feet are built around strings of syllables of four types: CVC, CV-CV, CVC-CV, and CV-CVC (exclusively at the right edge of the prosodic word).⁶

(40) *Quantity-Sensitive Footing*

a.		b.		c.		d.	
	(lèp)(pán)		(póle)		(lámba)		(léleŋ)
	‘stop by’		‘come’		‘go’		‘go about’

These trochees can be seen in many other places in the language: for instance, they are recruited by reduplicative affixes that select adjectives, verbs, and nouns (41a)-(41c).

(41) *The Moraic Trochee*

a.	(màʔ)-(màm)(mís), (CVC)-mammis RED-sweet	(màsi)-(másiŋ), (CVCV)-masiŋ RED-salty	(tʃàŋŋo)-(tʃáŋŋo), (CVCCV)-tʃaŋŋo RED-stupid	(màte)-(máte) (CVCV)-mate RED-dead
	‘A little sweet, a little salty, a little stupid, a little dead.’			
b.	(dàʔ)-(dàr)(ràs), (CVC)-darras RED-recite	(dùu)-(íúuŋ), (CVCV)-duduŋ RED-carry on head	(dùndu)-(íúndu), (CVCCV)-dundu RED-drink	(dùku)-(íúku) (CVCV)-duku RED-help
	‘Recite a little, carry back and forth, drink a little, help repeatedly.’			
c.	(dòʔ)-(dòt)(tòr), (CVC)-dottor RED-doctor	(dàtʃi)-(íátʃiŋ), (CVCV)-datʃiŋ RED-scale	(dàppa)-(íáppa), (CVCCV)-dappa RED-fathom	(dàli)-(íáli) (CVCV)-dali RED-earring
	‘Doctors, toy scales, a few fathoms, various kinds of earrings.’			

4.2 The Phonological Phrase

Above the prosodic word and beneath the intonational phrase, there is a single type of intermediate phonological domain that I will term the phonological phrase (Itô & Mester 2007, 2012, 2013; Elfner 2012; cf. Nespor & Vogel 1986; Hayes 1989). In Mandarin, its right edge is marked by a high tone that falls after low-level syntactic xps like noun-adjective and verb-adverb strings (42a)-(42b). In regular clauses, it also falls after arguments, phrasal adjuncts, the verbal complex, and the auxiliaries that host agreement (42c).

(42) The Phonological Phrase

- | | |
|--|--|
| <p>a. {_φ bojak kes sannal }^H
 bojaŋ kaiaŋ sanna^l
 house big very
 ‘A very big house’</p> | <p>b. {_φ bemme? towandi }^H
 bemme? towandi
 fall actually
 ‘Actually fell’</p> |
| <p>c. pure^H sita^H ɿotor^H dio^H
 pura i sita dottor dio
 PAST 3ABS meet doctor there
 ‘The doctors met there.’</p> | |

Phonological phrases are also built around many functional projections along the clausal spine in Mandarin (Brodskij, 2025a), yielding patterns of recursive phonological phrasing that mirror the syntax below (Selkirk 2009, 2011; Elfner 2012, 2015; Elordieta 2015; Van Handel 2021; see also Ladd 1986; Wagner 2005, 2010; Itô & Mester 2007, 2012, 2013; Ito & Mester 2021). Phrases of this type can be detected from a range of restrictions on segmental processes that emerge in positions medial in the intonational phrase. For instance, there is a process of Voiced Obstruent Lenition that reduces intervocalic /b d d̥ ʒ g/ to [w ɿ j ɣ]. In three-word strings of the order vso, this process applies at the left edges of the s and the o (43a). But in four-word vso strings where the o contains two prosodic words, it is blocked at the left edge of the o (43b). In the same

vein, it is always blocked at the left edges of final adjuncts to the *voiceP* (43a)-(43b).

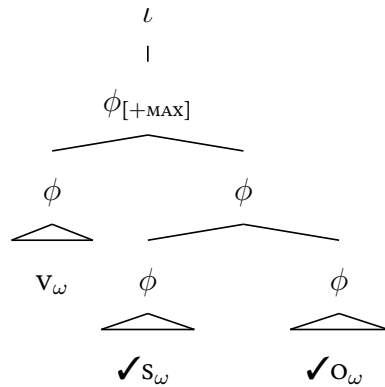
(43) *Voiced Obstruent Lenition*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | |
|-------------------|-------------------|--------------------|------------------|
| site ^H | ɣuru ^H | ɬotor ^H | dio ^H |
| sita i | guru | dottor | dio |
| meet 3ABS | teacher | doctor | there |
- ‘The teacher met the doctor there.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | |
|-------------------|-------------------|------------------------------|------------------|
| site ^H | ɣuru ^H | dottor malingao ^H | dio ^H |
| sita i | guru | dottor malingao | dio |
| meet 3ABS | teacher | doctor tall | there |
- ‘The teacher met the tall doctor there.’

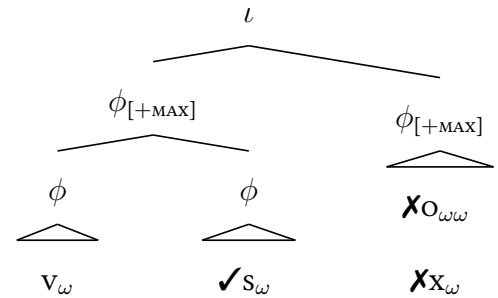
Brodkin 2025a argues that this restriction is keyed to the edge of the type of phonological phrase that is directly dominated by the intonational phrase: the *maximal* phrase phrase, in the terminology of Itô & Mester 2007, 2012, 2013. This analysis is sketched in (44): Voiced Obstruent Lenition proceeds between phonological phrases that are parsed into the same maximal phonological phrase but cannot cross over its edge.

(44) *The Maximal Phonological Phrase*

a. *Left-Edge Lenition*



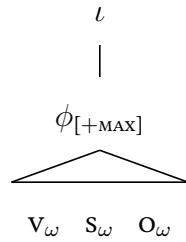
b. *Prosodic Blocking*



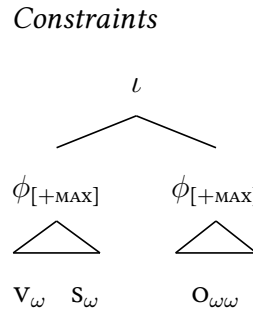
This analysis allows us to move forward with three facts in tow. First, the narrow *voiceP* is preferentially parsed into a large phonological phrase that contains the strings *vso*, *vsd*, and *vod* (where *D* = dative argument; (45a)). Second, the *voiceP* is split into multiple maximal phonological phrases when it contains more than three words (45b). Third, every adjunct to the *voiceP* is parsed into a maximal phonological phrase of its own (45c), in keeping with the ways that adjunction structures are treated at the interface (Brodkin 2025a; Hale & Selkirk 1987; Chen 1987; Truckenbrodt 1995, 1999).

(45) *Phonological Phrasing: Starting Rules*

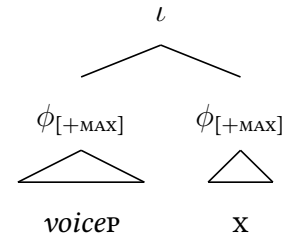
a. *VoiceP* → *Phrase*



b. *Weight*



c. *Adjunct Prosody*



4.3 The Positional Branching Constraint

In phrase-medial positions, the language allows prosodic words to end in monosyllabic feet. This parse emerges in words of the shape *cvc* (46a)-(46b) and words that end in the string *cvccvc* (46c)-(46d). Words that end in the strings *cvcv*, *cvccv*, and *cvcvc* host final disyllabic feet in these positions instead (46e)-(46f).

(46) *Phrase-Medial Metrical Structure*

a. (**tʃáʔ**) ma(méa)^H
 tʃaʔ mamea
 paint red
 ‘Red paint’

b. (**léŋ**) (ájú)^H
 leŋ adʒu
 glue wood
 ‘Wood glue’

- | | |
|---|---|
| <p>c. (dòt)(tór) ma(lìŋ)(gáo)^H
 dottor malingao
 doctor tall
 ‘Tall doctor’</p> | <p>d. ma(ɣàr)(rít) to(wándi)^H
 magarrin tobandi
 feverish actually
 ‘Actually feverish’</p> |
| <p>e. (gúru) ma(lìŋ)(gáo)^H
 guru malingao
 teacher tall
 ‘Tall teacher’</p> | <p>f. (mónɛʔ) to(wándi)^H
 monɛʔ tobandi
 sick actually
 ‘Actually sick’</p> |

At the right edge of the phonological phrase, monosyllabic feet are ruled out. We can see this effect in the phonological phrases that are medial in the intonational phrase but final in the maximal phonological phrase, like those built around arguments and predicates that fall before adjuncts to the *voiceP*. At the edges of those phrases, many repairs conspire to ensure that prosodic words will always end in disyllabic feet. Cvc roots, for instance, are rendered disyllabic by a process of ʔv-epenthesis that copies their vowels, infixes them into the word, and breaks hiatus with a glottal stop—yielding an effect segmentally similar to the “echo epenthesis” of the related Makassarese, Selayarese, and Konjo (Aronoff *et al.*, 1987; Friberg & Friberg, 1991; McCarthy, 1998; Gafos & Lombardi, 1999; Basri *et al.*, 1999; Kawahara, 2007; Stanton & Zukoff, 2018).

(47) *Echo Epenthesis*

- | | |
|---|--|
| <p>a. {ϕ[MAX] } {ϕ[MAX] }</p> <p>nalaŋa^H ǎʔ (tʃáʔaʔ)^H díó^H.
 na-alaŋaŋ aʔ tʃaʔ dio
 3ERG-get for 1ABS paint there
 ‘They got me some paint there.’</p> | <p>b. {ϕ[MAX] } {ϕ[MAX] }</p> <p>nalaŋa^H ǎʔ (léʔen)^H diyénaʔ^H.
 na-alaŋaŋ aʔ leŋ digenaʔ
 3ERG-get for 1ABS glue earlier
 ‘She got me some glue earlier.’</p> |
|---|--|

Final-cvccvc roots are forced to host the same types of feet, and to this end they undergo five analogous repairs. Final glottal stops delete in roots of the shape cvccv? (48a), medial codas delete in roots that host medial geminates, nasals, and glottal stops (48b)-(48d), and the final lateral deletes in the single root *sannal* ‘very’ (48e).

(48) *Segmental Deletions*

- a. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
 póle^H i(tʃíttʃi)^H djónɪŋ^H.
 pole i itʃíttʃi? dionɪŋ
 come 3ABS NAME yesterday
 ‘Chichi’ came yesterday.’
- b. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
 néte^H (iótɔr)^H diràmbónɪ^H.
 na-ita i dottor diruamboŋi
 3ERG-see 3ABS doctor a while ago
 ‘The doctor saw him a while ago.’
- c. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
 píto^H (páⁿdes)^H sàllíwaŋ^H.
 peita o pandeŋ sallibaŋ
 go look for 2ABS pineapple on the other side
 IDIOMATICALLY: ‘Go look for a wife in another village.’
- d. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
 píto^H (kúmil)^H sàllíwaŋ^H.
 peita o kuʔmil sallibaŋ
 go look for 2ABS jackfruit on the other side
 ‘Go look for jackfruit outside.’
- e. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
 bjase^H matʃoa (sánna)^H díó^H.
 bjasa i matʃoa sanna^l dio
 usually 3ABS good very there
 ‘It’s usually very good there.’

These effects reveal that the phonology requires the final foot in the phonological phrase to be the disyllabic trochee in (49). This is a top-down requirement for binarity (Booij 1999; Prieto 2005; Elordieta 2006), first documented in another corner of the language by Brodtkin 2025b, and I will refer to it as the Positional Branching Constraint.

(49) *The Positional Branching Constraint*



4.4 Syllable Attraction

In positions that are medial in the maximal phonological phrase, the Positional Branching Constraint is preferentially repaired in a different way. This second repair is linked to the metrical status of the syllable at the left edge of the following phonological phrase. As a result, there is a phonological difference between the two positions below: those that are followed by syllables that should be footed (50a) and those that are followed by syllables that should not (50b).

(50) *The Positional Divide*



In the first context, we see our usual repairs: before roots that are disyllabic (51a) and roots that begin with initial heavy syllables (51b), final-cvccvc roots show the usual patterns of reduction and cvc roots show the usual process of ?v-epenthesis.

(51) *Regular Repairs in the Maximal Phonological Phrase*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|------|-------------------------------|------------------------------|--------------------|
| néte ^H | | (ɪótor) ^H | (gúru) ^H | dío ^H . |
| na-ita | i | dottor | guru | dio |
| 3ERG-see | 3ABS | doctor | teacher | there |
- ‘The doctor saw the teacher there.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|------|-------------------------------|--|------------------------|
| wéŋa ^H | ĩ | (léʔek) ^H | (kàn)(díʔu) ^H | diyénaʔ ^H . |
| u-beŋaŋ | i | leŋ | kandiʔ-u | digenaʔ |
| 1ERG-give | 3ABS | glue | sibling-1GEN | earlier |
- ‘I gave some glue to my little sibling earlier.’

In the second position, we see two shifts. First, before trisyllabic roots that begin in light syllables, the phrase-final high tone shifts one syllable to the right. Second, the usual repairs cease: CVC roots show no epenthesis (52a), CVCCVC roots retain final glottal stops and all types of medial codas (52b)-(52d), and *sannal* retains its lateral (52f).

(52) *The Special Repair*

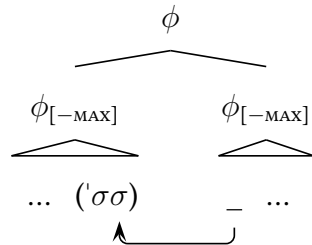
- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|------|------------|--|------------------------|
| wéŋa ^H | ĩ | lék | ka ^H (káʔu) ^H | diyénaʔ ^H . |
| u-beŋaŋ | i | leŋ | kakaʔ-u | digenaʔ |
| 1ERG-give | 3ABS | glue | sibling-1GEN | earlier |
- ‘I gave some glue to my older sibling earlier.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|------|--------------------|--|--------------------|
| néte ^H | | i(tʃit)tʃiʔ | gu ^H (rúna) ^H | dío ^H . |
| na-ita | i | itʃittʃiʔ | guru-na | dio |
| 3ERG-see | 3ABS | NAME | teacher-3GEN | there |
- ‘Chichi’ saw her teacher there.’
- c. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|------|---------------------------|--|--------------------|
| néte ^H | | (ɪòt) tór | gu ^H (rúʔu) ^H | dío ^H . |
| na-ita | i | dottor | guru-u | dio |
| 3ERG-see | 3ABS | doctor | teacher-1GEN | there |
- ‘The doctor saw my teacher there.’

- d. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | | |
|---------------------|---|-----------|-------------------------------------|--------------------|
| waláŋa ^H | ĩ | (pàn)dék | ka ^H (káʔu) ^H | dío ^H . |
| u-alan̄aŋ | i | panden̄ | kakaʔ-u | dio |
| 1ERG-get for 3ABS | | pineapple | sibling-1GEN | there |
- ‘I got some pineapple for my older sibling there.’
- e. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | | |
|---------------------|---|-----------|-------------------------------------|--------------------|
| waláŋa ^H | ĩ | kùʔmíl | ka ^H (káʔu) ^H | dío ^H . |
| u-alan̄aŋ | i | kuʔmil | kakaʔ-u | dio |
| 1ERG-get for 3ABS | | jackfruit | sibling-1GEN | there |
- ‘I got some jackfruit my older sibling there.’
- f. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | | |
|----------|------------------------------------|----------------------------|--|---------------------|
| n̄de | pa ^H lákam ^H | mat̄jóa (sàn)nál | d̄ʒe ^H (páʔna) ^H | dítĩ ^H . |
| ndan̄ i | palakan̄ | mat̄jóa sanna ^l | d̄ʒepaʔ-na | ditiŋ |
| not 3ABS | turns out | good very | tortilla-3GEN | there |
- ‘Turns out the tortillas weren’t very good there.’

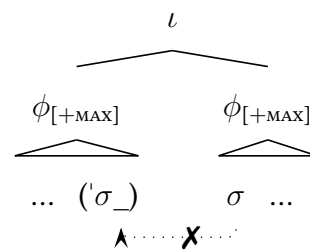
These effects reveal that the phonology preferentially constructs this disyllabic foot around roots of this shape by drawing in syllables that would otherwise go unfooted from the left edge of the following phonological phrase (53a). This is one of many processes in the phonology of the language that is blocked at the right edge of the maximal phonological phrase (53b), and its wider distribution has been documented by Brodtkin 2025b. I will refer to it as SYLLABLE ATTRACTION below.

(53) *Syllable Attraction*

a. *The Optimal Repair*



b. *The Top-Down Constraint*



4.5 Positional Licensing

At the right edge of the intonational phrase, we see a second effect: the language suspends every repair to the Positional Branching Constraint. At the edges of phrases in that position, cv(c) roots remain monosyllabic and cvccvc-final roots show no deletion.⁷

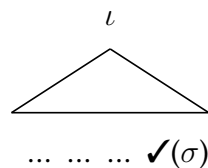
(54) *The Suspension Effect*

- | | |
|--|--|
| <p>a. {_ℓ naláŋa^H ãʔ (léŋ)^H }.</p> <p>na-aláŋaŋ aʔ leŋ</p> <p>3ERG-get for 1ABS glue</p> <p>‘She got me some glue.’</p> | <p>b. {_ℓ néte^H i(tŋít)(tŋíʔ)^H }.</p> <p>na-ita i itŋítŋíʔ</p> <p>3ERG-see 3ABS NAME</p> <p>‘Chichi’ saw him.’</p> |
|--|--|

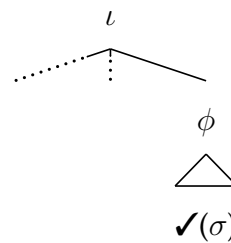
These effects trace a second top-down constraint: the intonational phrase preferentially hosts a final monosyllabic foot (55a) and supersedes the Positional Branching Constraint to license phrase-final monosyllabic feet in this position (55b).

(55) *Positional Licensing*

a. *The Final Foot*



b. *Suspending Minimality*



The case for this constraint emerges from a process of ʔ-epenthesis that operates at the right edge of the intonational phrase. Outside the right edge of that domain, words that contain odd numbers of light syllables typically receive the non-exhaustive parse in (56a). When those words surface at the right edge of the intonational phrase, however, they undergo a process of ʔ-epenthesis and receive the exhaustive parse in (56b). The same process never applies to roots that contain even numbers of light syllables, which are exhaustively parsed in the default case (56c)-(56d).

(56) *Glottal Epenthesis*

- | | |
|---|---|
| <p>a. {_ℓ démmo^H la(súna)^H áʔ^H }.</p> <p> dian mo lasuna a</p> <p> there is now shallot PRT</p> <p> ‘Now there are shallots, eh?’</p> | <p>b. {_ℓ démmo^H (làsu)(náʔ)^H }.</p> <p> dian mo lasuna</p> <p> there is now shallot</p> <p> ‘Now there are shallots.’</p> |
| <p>c. {_ℓ démmo^H (lémo)^H áʔ^H }.</p> <p> dian mo lemo a</p> <p> there is now citrus PRT</p> <p> ‘Now there are lemons, eh?’</p> | <p>d. {_ℓ démmo^H (lémo)^H }.</p> <p> dian mo lemo</p> <p> there is now citrus</p> <p> ‘Now there are lemons.’</p> |

In words of different shapes, this process applies whenever it can create a monosyllabic foot without derailing exhaustive footing through the rest of the word. We can see this effect in the behavior of three types of words that typically receive an exhaustive parse: those of the shapes CVC-CV, CVC-CV-CV, and CVC-CVC-CV. Outside of this edge, these words typically host final disyllabic feet—but at the right edge of the intonational phrase, they undergo ʔ-epenthesis to host the final monosyllabic feet in (57).

(57) *Final Monosyllabic Feet*

- | | |
|---|---|
| <p>a. {_ℓ démmo^H (ánde)^H áʔ^H }.</p> <p> dian mo ande a</p> <p> there is now food PRT</p> <p> ‘Now there’s food, eh?’</p> | <p>b. {_ℓ démmo^H (àn)(déʔ)^H }.</p> <p> dian mo ande</p> <p> there is now food</p> <p> ‘Now there’s food.’</p> |
| <p>c. {_ℓ démmo^H (àn)(d̥ʒoro)^H áʔ^H }.</p> <p> dian mo and̥ʒoro a</p> <p> there is now coconut PRT</p> <p> ‘Now there are coconuts, eh?’</p> | <p>d. {_ℓ démmo^H (ànd̥ʒo)(róʔ)^H }.</p> <p> dian mo and̥ʒoro</p> <p> there is now coconut</p> <p> ‘Now there are coconuts.’</p> |
| <p>e. {_ℓ démmo^H (àk)(kátta)^H áʔ^H }.</p> <p> dian mo akkatta a</p> <p> there is now plan PRT</p> <p> ‘Now there’s a plan, eh?’</p> | <p>f. {_ℓ démmo^H (àk)(kàt)(táʔ)^H }.</p> <p> dian mo akkatta</p> <p> there is now plan</p> <p> ‘Now there’s a plan.’</p> |

4.6 Reinforcers Revised

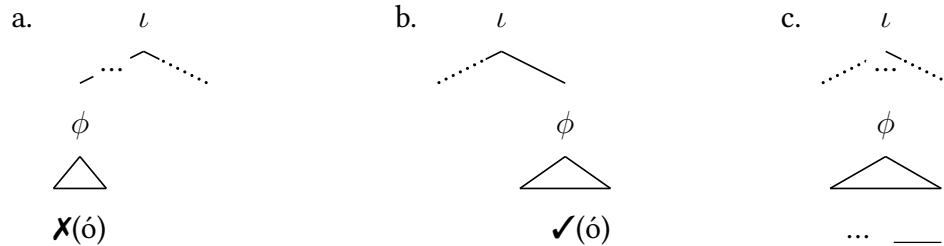
This footwork reveals that Reinforcer Postposing and Reinforcer Deletion have an optimizing effect. The reinforcers form words and phrases of their own: like possessors in SPEC,DP and phrasal adjuncts to the DP, they host word-level stress and carry phrase-final high tones. As a result, they implicitly violate the Positional Branching Constraint: the reinforcers form phonological phrases and cannot host disyllabic feet.

(58) *The Reinforcers: Phrase-Final Monosyllabic Feet*

- | | |
|--|---|
| <p>a. do wúku^H díó^H (óʔ)^H
 do buku dio o
 that book there there
 ‘That book there’</p> | <p>b. de wukúna^H ikátʃoʔ^H (éʔ)^H
 de buku-na ikátʃoʔ e
 this book-3GEN NAME here
 ‘This book here of Kacho’s.’</p> |
|--|---|

As a result, we can understand Reinforcer Postposing and Reinforcer Deletion as output-oriented repairs to the system of Positional Binarity: Reinforcer Postposing draws the reinforcers to positions that license phrase-final monosyllabic feet (59b) and Reinforcer Deletion suppresses them whenever Postposing cannot apply (59c).

(59) *Reinforcers and Positional Minimality*



To cement this analysis, we can now show that both repairs alternate transparently with Syllable Attraction. The case begins with a brief foray into the system of weight. When we introduce additional prosodic words into arguments that are medial in the *voiceP*, the verb and the first argument are always parsed into a maximal phonological

phrase that excludes all material that follows (Brodkin, 2025a). When we add additional prosodic words to the verbal complex, however, we can force it to form a maximal phonological phrase of its own—and in this context, two-word arguments that are medial in the *voiceP* will begin to be parsed into maximal phonological phrases with one-word arguments to their right. We can see this shift in vso strings in the system of Syllable Attraction: when the subject contains two words, it typically cannot force Syllable Attraction into the o (60a), but when the verb is made heavier, it can (60b).

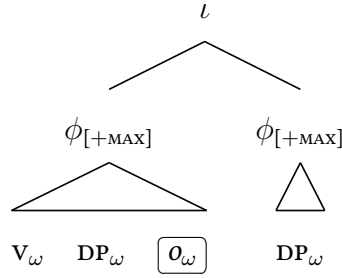
(60) *The Weight Constraint*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | |
|-------------------|------|------|----------------------|-----------------------|------------------------|
| wéŋa ^H | ĩ | wúku | (kétŋu) ^H | gu(rúʔu) ^H | diyénaʔ ^H . |
| u-beŋaŋ | i | buku | kettŋuʔ | guru-u | digenaʔ |
| 1ERG-give | 3ABS | book | little | teacher-1GEN | earlier |
- ‘I gave a little book to my teacher earlier.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | | |
|------|--------------------|------|------|-----------|-------------------------------------|------------------------|
| mé | wéŋaŋ ^H | i | búku | ket(tŋúʔ) | gu ^H (rúʔu) ^H | diyénaʔ ^H . |
| mane | u-beŋaŋ | i | buku | kettŋuʔ | guru-u | digenaʔ |
| just | 1ERG-give | 3ABS | book | little | teacher-1GEN | earlier |
- ‘I gave a little book to my teacher earlier.’

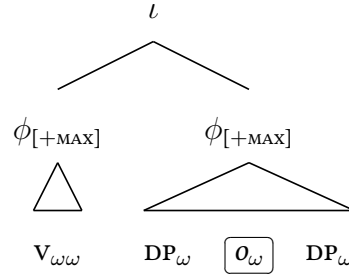
This footwork sets up the following prediction. When the verbal complex contains a single word and a reinforcer is linearized after a one-word argument in the *voiceP*, we should expect to see the parse in (61a): as the reinforcer forms a prosodic word, the phonology will not build a maximal phonological phrase around the string V-S-REINFORCER-O. When the verbal complex contains two words, however, we should see the shift in (61b): the phonology should build a maximal phonological phrase around the string S-REINFORCER-O. In that second context, Syllable Attraction should occur.

(61) *The Weight Constraint*

a. *Two-Word Subjects*



b. *The Manipulation*



This prediction is correct. When the reinforcers are initially linearized into positions that would force them to fall at the right edges of maximal phonological phrases if they remained in-situ, they move or delete (62a). When they are linearized into positions that should allow Syllable Attraction, however, they surface in place (62b).

(62) *Reinforcers Trigger Syllable Attraction*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | | | |
|-------------------|------|------|-------------------|---|---------------------|----------------------|-------------------|
| wéŋa ^H | ĩ | io | wúku ^H | — | yurúʔu ^H | diyénaʔ ^H | óʔ ^H . |
| u-beŋaŋ | i | do | buku | | guru-u | digenaʔ | o |
| 1ERG-give | 3ABS | that | book | | teacher-1GEN | earlier | there |
- ‘I gave that book there to my teacher earlier.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | | | |
|------|--------------------|------|------|-------------------|-------|--------------------------------------|------------------------|
| mé | wéŋaŋ ^H | i | do | wúku ^H | (ó | ɣu) ^H (rúʔu) ^H | diyénaʔ ^H . |
| mane | u-beŋaŋ | i | do | buku | o | guru-u | digenaʔ |
| just | 1ERG-give | 3ABS | that | book | there | teacher-1GEN | earlier |
- ‘I just gave that book there to my teacher earlier.’

With the same manipulations, we can force the reinforcers to follow their associates in any position that allows them to force Syllable Attraction. In complex DPs, for instance, we can force them to surface immediately after their associates by introducing rightward specifiers that contain exactly one word and begin with unstressed syllables (63a). In the left periphery, in turn, we can force them to follow associates in the same positions that host WH-phrases and allow Syllable Attraction into the verb (63b).⁸

(63) *Reinforcers in-Situ*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
do wukúna^H (ó yu)^H(rúʔu)^H dó ɿ méja^H.
do buku-na o guru-u dío dí med̥ʒa
that book-3GEN there teacher-1GEN there on table
‘That book of my teacher’s there on the table.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
do wúku^H (ó na)^H(wémme)^H maníni^H.
do buku o na-bemme? manini
that book there will-fall later
‘That book there will fall later.’

In these positions, in turn, we can see that the reinforcers are freely crossed by other reinforcers that are morphologically linearized further to the left. When the reinforcers can trigger Syllable Attraction and remain in-situ, in other words, they are phonologically deactivated and cease to intervene in the system of relative locality.

(64) *Phonological Deactivation*

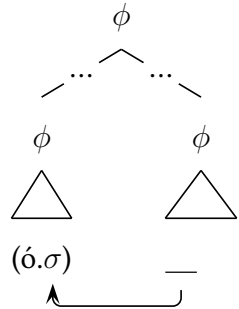
- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
de wúku kəs sànnálna^H — do solána^H (ó i)^H(kátʃoʔ)^H (éʔ)^H.
de buku kajaɲ sanna¹-na do sola-na o ikatʃoʔ e
this book big very-3GEN that friend-3GEN there NAME here
‘This really big book here of that friend there of Kacho’s.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
mé nawéɲa^H ɿ ɿe sànaéke^H — do wúku^H (ó yu)^H(rúna)^H (éʔ)^H.
mane na-beɲaɲ i de sanaeke do buku o guru-na e
just 3ERG-give 3ABS this kid that book there teacher-3GEN here
‘This kid here just gave that book there to his teacher.’

The result is a decisive case that Reinforcer Postposing and Reinforcer Deletion form part of the system of Positional Minimality: as they are bled by Syllable Attraction, they must occur in the phonology and be forced by the Positional Branching Constraint. We can thus understand them as a ranked series of repairs to rescue the reinforcers from

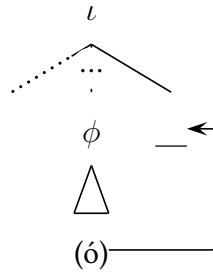
their internal tension: when possible, they trigger Syllable Attraction (65a), when they cannot, they shift (65b), and when that repair is ruled out, they cease to be expounded (65c).

(65) *Item-Specific Repairs*

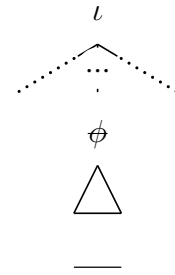
a. *First Choice*



b. *Second Choice*



c. *Third Choice*



4.7 Suspended Exponence

To round out our account of Reinforcer Deletion, we can conclude by integrating these effects with the wider set of item-specific repairs to the Positional Branching Constraint. The first of these is a type of suspended exponence that emerges around a verbal suffix: the transitivity marker *-í*. This suffix is overtly expounded in phrase-medial positions (66a), at the right edge of the intonational phrase (66b), and in the phrase-final positions that allow Syllable Attraction (66c).

(66) *The Transitivity Suffix*

- a. {_ϕ wóloʔ(i) towándi }^H {_ϕ déʔ }^H.
u-oloʔ-i tobandi de
1ERG-like-I actually.3ABS I guess

‘I actually like it, I guess.’

- b. {_ℓ bése^H palákam^H wóloʔ(iʔ)^H }.
biasa i palakanj u-oloʔ-i
usually 3ABS turns out 1ERG-like-I

‘Turns out I usually like it.’

- c. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|----------------------|-------------|---------------------------------------|--------------------|
| bése ^H | palákam ^H | wóloʔ(i | je) ^H (páʔna) ^H | dío ^H . |
| biasa i | palakanj | u-oloʔ-i | dʒepaʔ-na | dio |
| usually 3ABS | turns out | 1ERG-like-I | tortilla-3GEN | there |
- ‘Turns out I usually like their tortillas there.’

In the contexts that prohibit monosyllabic feet and Syllable Attraction, this suffix is not expounded. As a result, it disappears at the edges of maximal phonological phrases (67a) and non-maximal phonological phrases that precede footed syllables (67b).⁹

(67) *Suspended Exponence*

- a. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | |
|-------------------|----------------------|--------------------------|--------------------|
| bése ^H | palákam ^H | wóloʔ^H | dío ^H . |
| biasa i | palakanj | u-oloʔ-i | dio |
| usually 3ABS | turns out | 1ERG-like-I | there |
- ‘Turns out I usually like it there.’
- b. $\{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\} \quad \{\phi_{[MAX]}\}$
- | | | | | |
|-------------------|----------------------|-------------------------|---------------------|--------------------|
| bése ^H | palákam ^H | wólo^H | (ánde) ^H | dío ^H . |
| biasa i | palakanj | u-oloʔ-i | ande | dio |
| usually 3ABS | turns out | 1ERG-like-I | food | there |
- ‘Turns out I usually like the food there.’

The same system drives irregular alternations in the negator *ndang* and the WH-words *a*, *na*, *nai* ‘what, which, who’. In the positions that allow monosyllabic feet or Syllable Attraction, these elements take their unmarked forms: Brodkin 2025b notes that $\sqrt{\text{WHAT}}$ is realized as *a* in the contexts that allow Syllable Attraction (68a), and here we can note that at the edge of the intonational phrase the negator is realized as *ndang* (68b).

(68) *Monosyllabic Allomorphs*

- a. $\{\phi_{[MAX]}\}$
- | | |
|----------------|--|
| (á | na) ^H wàluwánj ^H ? |
| a | na-baluanj |
| what 3ERG-sell | |
- ‘What’s he selling?’

- b. $\widehat{itfittj}^H$ $\widehat{lulus}^H$, $\{ \iota$ $\widehat{tapi}$ $\widehat{ikatfo}^H$ **($\widehat{ndanj}$)^H** $\}$.
 $\widehat{itfittj}^?$ lulus tapi $\widehat{ikatfo}^?$ ndan
NAME graduate but NAME not *voiceP*
‘Chichi’ graduated, but Kacho’ didn’t.’

In the contexts that prohibit monosyllabic feet and Syllable Attraction, Brodkin 2025b observes, these heads supplete. At the right edge of the maximal phonological phrase, the negator *ndang* takes the form *andiang* (69b), and at the edges of non-maximal phonological phrases that fall before footed syllables, $\sqrt{\text{WHAT}}$ takes the form *apa* (69a).

(69) *Polysyllabic Allomorphs*

- a. $\{\phi_{[MAX]}\}$
($\widehat{apa}$)^H ($\widehat{pil}$) $\widehat{lowa}^H$?
a pin-loa
what make noise
‘What’s ringing?’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
 $\widehat{itfittj}^H$ $\widehat{lulus}^H$, tapi ιo $\widehat{kátfo}^H$ **($\widehat{an}$)($\widehat{dianj}$)^H** $\widehat{ó}^H$.
 $\widehat{itfittj}^?$ lulus tapi do $\widehat{kátfo}^?$ ndan o
NAME graduate but that NAME not Δ there
‘Chichi’ graduated, but that Kacho’ there didn’t.’

The following diagrams illustrate these effects. The suffix *-í* and the four suppling four heads are expounded normally in the positions that allow them to head monosyllabic feet, like phrase-medial positions and the right edges of the intonational phrase:

(70) *Regular Exponence*

- a. ι

 ϕ
 $\triangle$
(á)
- b. ι

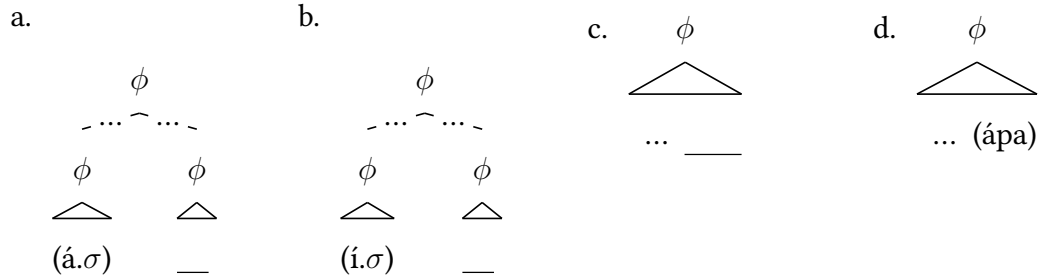
 ϕ
 $\triangle$
(í)
- c. ϕ

... (á) ...
- d. ϕ

... (í) ...

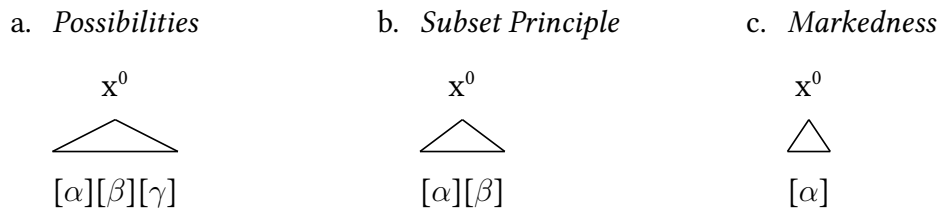
But at the right edge of the phonological phrase, we see the following alternation: when these heads can trigger Syllable Attraction, they are expounded in their usual way, and when they cannot, *-i* is not expounded and the remaining heads supplete.

(71) *Disruptions to Exponence*



To capture these effects, I propose that exponence proceeds through a two-step process that is distributed between the morphology and the phonology. Following the standard view in Distributed Morphology, I take morphological exponent selection to occur in a cyclic and bottom-up process that targets terminal nodes in a series of steps that follow impoverishment (Halle & Marantz, 1993), node insertion and deletion, fission and fusion (Hewett, 2023), and linearization (Adger 2006; ?; Ostrove 2018; see also Embick & Noyer 2001). I assume that it proceeds in three steps that I will refer to collectively as the operation MATCH. First, it identifies the complete set of exponents that can be matched with the target head. Second, it narrows down the set of possible exponents on the basis of the subset principle (Halle & Marantz, 1993), which favors the candidates that most closely match the featural specification and position of the target (Bobaljik, 2000, 2012; Embick, 2010; Gouskova & Bobaljik, 2020). Third, when multiple candidates are favored to the same degree, it narrows the set further on the basis of morphological markedness constraints (Arregi & Nevins, 2012).

(72) *Stage One: Match*



I propose that exponents are inserted into terminal nodes by a second process, INSERT, which occurs in the phonology and proceeds in parallel with the translation of constituent structure and linear order. In the simplest cases, these systems translate linearized and unexponed morphological representations like (73a) to linearized and exponed prosodic representations like (73b). But in contexts of conflict with output constraints, I propose that the phonology can force the type of suspended exponence in (73c): one that directly suspends insertion to satisfy output constraints (and see also Raffelsiefen 1999, 2004; Kurisu 2001; Henderson 2012, and Aissen 2017).

(73) *Stage Two: Insert*

a. *Morphology*

XP
|
x⁰
△
[α]

b. *Phonology*

ϕ
|
ω
△
[α]

c. *Suspension*

ϕ
|
⊘
△
—

5 Linearization and Exponence at the Interface

The final task of the paper is to integrate this analysis with the machinery of Match Theory (Selkirk, 2009, 2011), the leading theory of syntax-prosody translation (Ishihara & Kalivoda 2022). This theory situates the transition of constituent structure in a parallel and global phonological calculus that weighs output constraints against ranked and violable pressures for isomorphism which map morphosyntactic words to prosodic words and morphosyntactic xps to phonological phrases. In this calculus, I propose that the linearization and allomorph selection of the morphology are also guarded by ranked and violable constraints—allowing them to be derailed by output constraints.

5.1 Relinearization

To capture the facts of relinearization, I propose that the linear order of the morphology is guarded by the isomorphism constraint in (74), which checks the linear order of pairs of terminal nodes in the input and compares it against the order of their exponents in the output (and see also Halpern 1995; Göbbel 2007; López 2009a; Bennett *et al.* 2016; Aissen 2017; Clemens & Coon 2018; Clemens 2019; Kusmer 2019; Potsdam 2022).

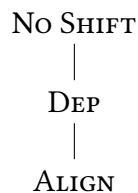
(74) NO SHIFT

Assign one violation for every pair of constituents α, β , such that α is linearized before β in the morphological input, for which the output correspondent of α is not linearized before the output correspondent of β in the phonological output.

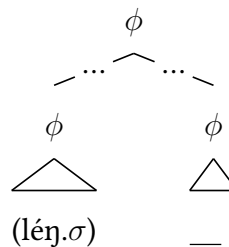
The repairs to the Positional Branching Constraint emerge from the relative ranking of NO SHIFT with other constraints. The behavior of the cvc roots follows from the interaction of NOSHIFT with DEP, which prevents epenthesis, and the ALIGN constraints, which force isomorphism between morphosyntactic and prosodic domains (McCarthy & Prince 1993; Selkirk 1995a,b). With the ranking in (75a), we can force them to trigger Syllable Attraction when they can (75b) and show epenthesis when they cannot (75c).

(75) *Syllable Attraction*

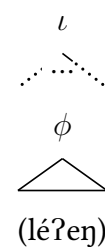
a. *Ranking*



b. *Attraction*



c. *Epenthesis*



To force Reinforcer Postposing, in turn, we can introduce two further constraints. The first is the lexically indexed version of DEP in (76a), which prevents ʔv-epenthesis in the reinforcers (for discussion: Pater 2000, 2006, 2008; Gouskova 2012). The second is the

TETHERING constraint in (76b), which prevents the reinforcers from shifting backwards to the left whenever it would provide a shorter path toward the right edge of an intonational phrase. I take this to be an isomorphism constraint that guides the translation of selection, forcing the reinforcers to fall into the intonational phrases that host their selecting demonstratives (see also Clemens & Coon 2018; Clemens 2019).

(76) *Reinforcers: Constraints*

a. DEP- $V_{e/o}$

Assign one violation for every vocalic segment in the output that has no correspondent in the input in the exponents of $\sqrt{\text{HERE}}$ and $\sqrt{\text{THERE}}$.

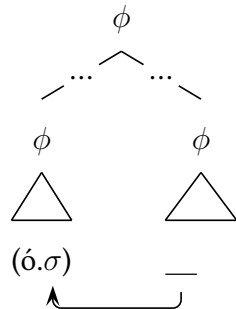
b. TETHERING

Assign one violation for every demonstrative-reinforcer pair α, β , such that α selects β in the morphological input, for which the output correspondents of α and β are not in the same intonational phrase in the phonological output.

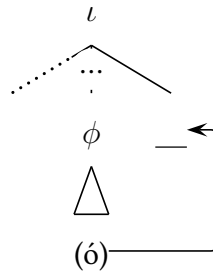
By the ranking these two constraints above NO SHIFT and ALIGN, we can ensure that the reinforcers will trigger Syllable Attraction whenever they can (77a), move when they cannot (77b), and never escape the intonational phrase (77c).

(77) *Reinforcer Postposing*

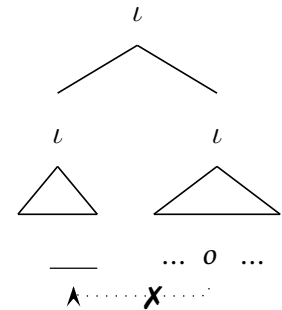
a. *Attraction*



b. *Postposing*



c. *Tethering*



To derive the path of Reinforcer Postposing, at last, we can introduce two types of

MATCH constraint (and see especially (Elfner, 2012, 2015; Bennett *et al.*, 2016)). The first is the MAX constraint in (78a), which forces the construction of phonological phrases that contain the exponents of all the terminal nodes in their corresponding xps. The second is the DEP constraint in (78b), which prevents the formation of phonological phrases that contain the exponents of terminal nodes not contained in their corresponding xps.

(78) *Match Constraints*

a. MATCH-XP

Assign one violation for every XP that dominates a set of terminal nodes in the morphological input that does not correspond to a phonological phrase that contains the output correspondents of all of those terminal nodes in the phonological output.

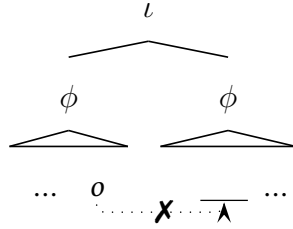
b. MATCH- ϕ

Assign one violation for every phonological phrase in the phonological output that contain the output correspondents of terminal nodes not contained in its corresponding XP in the morphological input.

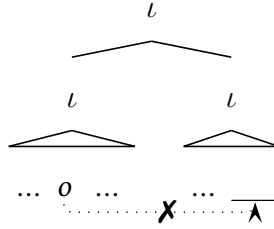
These constraints allow us to force the reinforcers to shift to the right edge of the intonational phrase and form maximal phonological phrases of their own. By ranking NoSHIFT over the MAX constraint, to begin, we can force the reinforcers to shift when they have no other escape from the Positional Branching Constraint. By ranking DEP over NoSHIFT, in turn, we can prevent them from simply shifting into the linearly closest position that should allow Syllable Attraction (79a). From this position, at last, we can derive two further results from violation profile of NoSHIFT itself: as this constraint imposes gradient violations that are sensitive to linear order, it will always force movement to the edge of the smallest containing intonational phrase (79b) and force movement of the needy reinforcers that are linearized furthest to the right (79c).¹⁰

(79) *Deriving the Path*

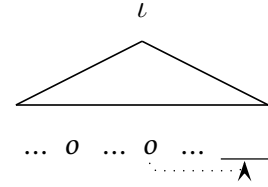
a. *Absolute Locality*



b. *Absolute Locality*



c. *Relative Locality*



5.2 Suspended Exponence

To capture the pattern of Reinforcer Deletion, I propose that INSERT is guided by three types of isomorphism constraints. The first is the IDENT constraint in (80a), which forces the phonology to preserve lexical hierarchies of markedness: when the morphology allows multiple exponents to be inserted into a particular node, this constraint forces insertion of the exponent that is lexically unmarked (Tranel 1996; Rubach & Booij 2001; Klein 2003, and especially Mascaró 1996a,c, 2007). The second is the IDENT constraint in (80b), which forces the phonology to insert the allomorphs selected by the morphology (cf. Bonet *et al.* 2007). The third is the MAX constraint in (80c), which forces insertion to occur (and see also Raffelsiefen 1996, 1999, 2004; Kurisu 2001; Henderson 2012).

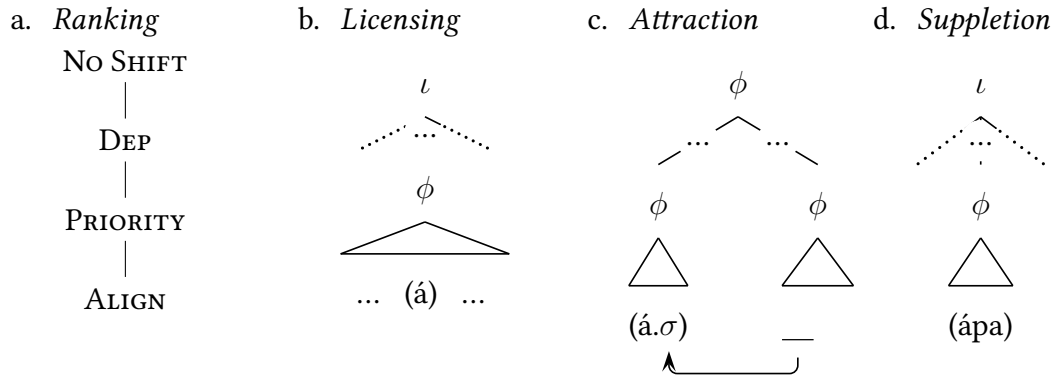
(80) *Isomorphism Constraints: Exponence*

- a. PRIORITY: Given an input containing an x^0 that can be realized with the lexically ordered allomorphs $\{M_1, M_2 \dots M_n\}$ and a candidate that contains M_i , assign as many violation marks as the depth of ordering between M_i and M_1 .
- b. RESPECT: Assign one violation for every terminal node α in the morphological input such that MATCH selects the exponent β for which β is not the output correspondent of α in the phonological output.

- c. EXPONE: Assign one violation for every terminal node in the input that is not targeted by INSERT to receive a correspondent in the output.

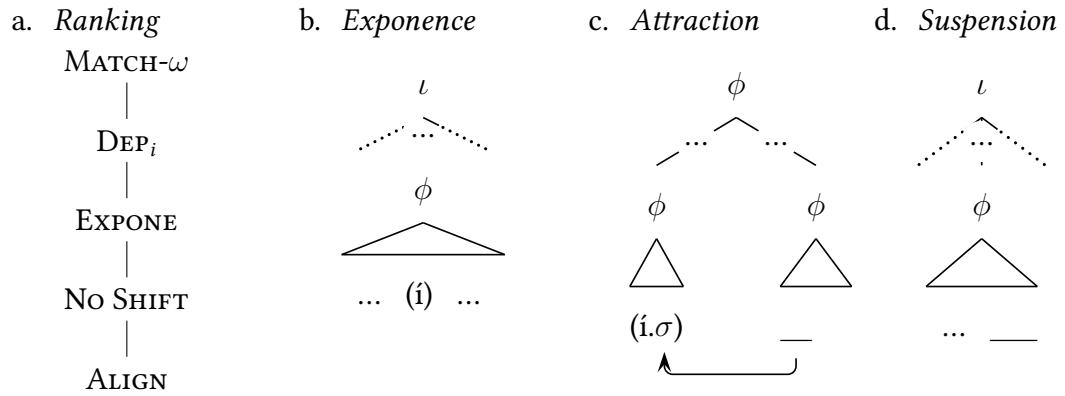
The patterns of suppletion in this system emerge from the ranking in (81a): taking PRIORITY to systematically favor the insertion of monosyllabic allomorphs, we can leverage this constraint to force them out in the contexts that allow them to appear (81b). We can then rank PRIORITY over ALIGN to force them in the contexts that allow Syllable Attraction (81c) and rank PRIORITY beneath DEP to force the marked allomorphs to emerge at the right edges of the phonological phrases that prohibit this repair (81d).

(81) *Suppletion*



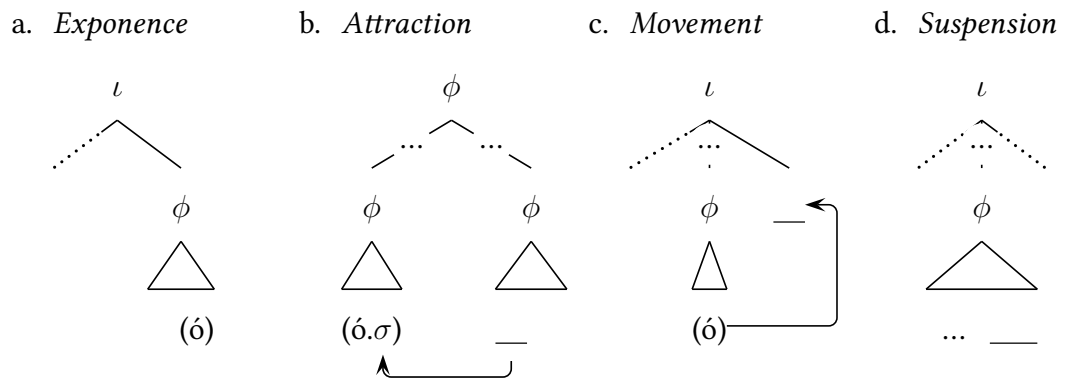
Suspended exponence of the transitivity suffix emerges from the ranking in (82a). When the morphological process of MATCH selects the exponent *i* for this head, the phonology inserts it in the contexts that allow monosyllabic feet (82b). In the contexts that allow Syllable Attraction, the ranking of EXPONE over ALIGN forces the same result (82c). In other contexts, the ranking of the Positional Minimal Constraint and an item-specific DEP constraint over EXPONE and NoSHIFT forces another repair. I assume that movement is ruled out in this context by higher-ranking MATCH constraints that operate at the level of the word and prevent *-i* from being parsed into any other prosodic word. As a result, I take these constraints to force the type of non-exponence in (82d).

(82) *Suspended Exponence*



Reinforcer Deletion emerges in exactly the same way. EXPONE forces the reinforcers to be realized in the contexts that allow monosyllabic feet (83a). Whenever Syllable Attraction is possible, the ranking of EXPONE and NO SHIFT over ALIGN forces that repair (83b). When Syllable Attraction is impossible, in turn, the ranking of EXPONE over NO SHIFT forces Postposing (83c). When Postposing cannot occur, at last, the ranking of the item-specific DEP constraint over EXPONE forces the suspension of exponence in (83d).

(83) Reinforcer Deletion



6 Conclusion

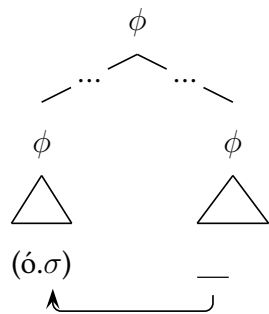
The result is a case that the systems of linearization and exponence must be split into two distinct stages and distributed across the morphological module and the phonology.

Empirically, this conclusion allows us to reinforce and extend the results of the wide literature on output-optimizing patterns of phonological displacement and exponence. Theoretically, in turn, it lends support to the hypothesis that the representations of the morphosyntax are translated into phonological terms in through an output-optimizing and imperfect translation. Conceptually, at last, they suggest that we might extend a similar treatment to other phenomena subsumed under the label of “PF effects”—and revisit the wider balance of labor between the morphology and the phonology.

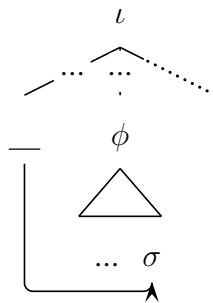
To conclude our investigation, I would like to work back to the theory of movement itself. From the theoretical perspective developed here, it seems clear that the phonology can force relinearization in response to pressures of three types: constraints that force movement into particular positions, like the pressures behind Syllable Attraction (84a), constraints that force movement out of particular positions, like the pressures that shunt minimal elements into second position (84b), and pressures internal to moving elements themselves, like the tension that forces the reinforcers to postpose (84c).

(84) *Phonological Relinearization: Three Types*

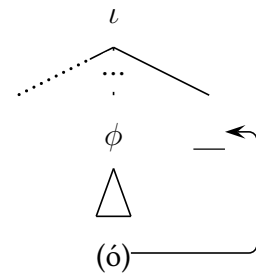
a. *Attraction*



b. *Push*



c. *Greed*



The result is that there are phonological counterparts to all three forces that have been argued to drive displacement in the syntax: ATTRACT (Chomsky, 2001), PUSH constraints (Diesing, 1992; Moro, 2000; Stroik, 2009; Chomsky, 2013), and GREED (Chomsky, 1995; Bošković, 1995). Future work will tell what this connection may reveal.

References

- Adger, David. 2006. Post-syntactic movement and the Old Irish verb. *Natural Language & Linguistic Theory*, 24(3), 605–654.
- Agbayani, Brian, & Golston, Chris. 2010. Phonological movement in classical Greek. *Language*, 86(1), 133–167.
- Agbayani, Brian, & Golston, Chris. 2016. Phonological constituents and their movement in Latin. *Phonology*, 33(1), 1–42.
- Agbayani, Brian, Golston, Chris, & Ishii, Toru. 2015. Syntactic and prosodic scrambling in Japanese. *Natural Language & Linguistic Theory*, 33(1), 47–77.
- Aissen, Judith. 2017. Special clitics and the right periphery in Tsotsil. *On looking into words (and beyond)*, 235.
- Anderson, Stephen R. 2008. English reduced auxiliaries really are simple clitics. *Lingue e linguaggio*, 7(2), 169–186.
- Aronoff, Mark, Arsyad, Azhar, Basri, Hasan, & Broselow, Ellen. 1987. Tier configuration in Makassarese reduplication. *Pages 1–15 of: Bosch, A., Need, B., & Schiller, E. (eds), Papers from the 23rd Annual Regional Meeting of the Chicago Linguistic Society, Part II: Parasession on Autosegmental and Metrical Phonology*. Chicago, IL: University of Chicago.
- Arregi, Karlos, & Nevins, Andrew. 2012. *Morphotactics: Basque auxiliaries and the structure of spellout*. Vol. 86. Springer Science & Business Media.
- Aylett, Matthew, & Turk, Alice. 2004. The smooth signal redundancy hypothesis: A functional explanation for relationships between redundancy, prosodic prominence, and duration in spontaneous speech. *Language and speech*, 47(1), 31–56.
- Basri, Hasan. 1999. *Phonological and syntactic reflections of the morphological structure of Selayarese*. Ph.D. thesis, State University of New York at Stony Brook.
- Basri, Hasan, Broselow, Ellen, & Finer, Daniel. 1999. Clitics and crisp edges in Makassarese. *Toronto Working Papers in Linguistics*, 16(2).
- Basri, Hasan, Broselow, Ellen, & Finer, Daniel. 2012. The end of the word in Makassar languages. *Pages 111–147 of: Borowsky, Toni, Kawahara, Shigeto, Shinya, Takahito, & Sugahara, Mariko (eds), Prosody matters: Essays in honor of Lisa Selkirk*. Sheffield: Equinox.
- Béjar, Susana. 1999. Agreement alternations and functional licensing in Selayarese. *Toronto Working Papers in Linguistics*, 16(2).
- Bennett, Ryan, & Elfner, Emily. 2019. The syntax–prosody interface. *Annual Review of Linguistics*, 5, 151–171.
- Bennett, Ryan, Elfner, Emily, & McCloskey, James. 2016. Lightest to the right: An apparently anomalous displacement in Irish. *Linguistic Inquiry*, 47(2), 169–234.
- Bernstein, Judy B. 1997. Demonstratives and reinforcers in Romance and Germanic languages. *Lingua*, 102(2-3), 87–113.

- Bobaljik, Jonathan David. 2000. The ins and outs of contextual allomorphy. *University of Maryland working papers in linguistics*, **10**, 35–71.
- Bobaljik, Jonathan David. 2012. *Universals in comparative morphology: Suppletion, superlatives, and the structure of words*. MIT Press.
- Bonet, Eulàlia, Lloret, Maria-Rosa, & Mascaró, Joan. 2007. Allomorph selection and lexical preferences: Two case studies. *Lingua*, **117**(6), 903–927.
- Booij, Geert. 1998. Phonological output constraints in morphology. *Phonology and morphology of the Germanic languages*, **143163**.
- Booij, Geert. 1999. *The phonology of Dutch*. Oxford University Press.
- Bošković, Željko. 1995. Case properties of clauses and the Greed Principle. *Studia Linguistica*, **49**(1), 32–53.
- Brodkin, Dan. 2025a. The Prosody of the Extended VP. *Natural Language and Linguistic Theory*.
- Brodkin, Dan. 2025b. Suppletion in Global Perspective. *Linguistic Inquiry*.
- Brodkin, Dan. 2026. Demonstratives, Reinforcers, and Rightward Movement: Phonological Displacement in Mandarin. *Linguistic Inquiry*.
- Bye, Patrik, & Svenonius, Peter. 2012. Non-concatenative morphology as epiphenomenon1. *The morphology and phonology of exponence*, 427–495.
- Chen, Matthew Y. 1987. The syntax of Xiamen tone sandhi. *Phonology*, **4**, 109–149.
- Chomsky, Noam. 1995. *The Minimalist Program*. MIT press.
- Chomsky, Noam. 2001. Derivation by phase. *Pages 1–52 of: Kenstowicz, M. (ed), Ken Hale: a life in language*. MIT Press.
- Chomsky, Noam. 2013. Problems of projection. *Lingua*, **130**, 33–49.
- Clemens, Lauren. 2019. Prosodic noun incorporation: The relationship between prosody and argument structure in Niuean. *Syntax*, **22**(4), 337–377.
- Clemens, Lauren, & Coon, Jessica. 2018. Deriving verb-initial word order in Mayan. *Language*, **94**(2), 237–280.
- De Belder, Marijke. 2020. A split approach to the selection of allomorphs: Vowel length alternating allomorphy in Dutch. *Glossa: a journal of general linguistics*, **5**(1).
- Diesing, Molly. 1992. *Indefinites*. Cambridge, MA: MIT press.
- Diesing, Molly, & Zec, Draga. 2011. Interface effects: Serbian clitics. *Experiments at the interfaces, syntax and semantics*, **37**, 1–30.
- Diesing, Molly, & Zec, Draga. 2017. Getting in the first word: Prosody and predicate initial sentences in Serbian. *Glossa: a journal of general linguistics*, **2**(1).
- Elfner, Emily. 2015. Recursion in prosodic phrasing: Evidence from Connemara Irish. *Natural Language & Linguistic Theory*, **33**(4), 1169–1208.

- Elfner, Emily Jane. 2012. *Syntax-prosody interactions in Irish*. University of Massachusetts Amherst.
- Elordieta, Gorka. 2006. Binariness at the left edge of Intonational Phrases. In: *GLOW Workshop on Prosodic Phrasing*.
- Elordieta, Gorka. 2015. Recursive phonological phrasing in Basque. *Phonology*, 32(1), 49–78.
- Embick, David. 2000. Features, syntax, and categories in the Latin perfect. *Linguistic inquiry*, 31(2), 185–230.
- Embick, David. 2007. Linearization and local dislocation: Derivational mechanics and interactions. *Linguistic analysis*, 33(3-4), 303–336.
- Embick, David. 2010. *Localism versus globalism in morphology and phonology*. Vol. 60. MIT Press.
- Embick, David, & Noyer, Rolf. 2001. Movement operations after syntax. *Linguistic inquiry*, 32(4), 555–595.
- Esser, S.J. 1938. *Atlas van tropisch Nederland*. Amsterdam: Nederlandsch Aardrijkskundig Genootschap.
- Féry, Caroline. 2013. Focus as prosodic alignment. *Natural Language & Linguistic Theory*, 31(3), 683–734.
- Féry, Caroline. 2017. *Intonation and prosodic structure*. Cambridge University Press.
- Féry, Caroline, & Ishihara, Shinichiro (eds). 2016. *The Oxford handbook of information structure*. Oxford University Press.
- Finer, Daniel. 1999. Cyclic clitics in Selayarese. *Toronto Working Papers in Linguistics*, 16(2).
- Finer, Daniel L. 1997. Contrasting $\bar{A}$ -dependencies in Selayarese. *Natural Language & Linguistic Theory*, 15(4), 677–728.
- Finer, Daniel L. 1998. Sulawesi relatives, V-raising, and the CP-complement hypothesis. *Canadian Journal of Linguistics/Revue canadienne de linguistique*, 43(3-4), 283–306.
- Friberg, Barbara. 1991. Ergativity, focus and verb morphology in several South Sulawesi languages. *Vical*, 2, 103–130.
- Friberg, Barbara. 1996. Konjo's peripatetic person markers. *Pacific Linguistics. Series A. Occasional Papers*, 84, 137–171.
- Friberg, Barbara, & Jerniati. 2000. *Mandar Conversations*. Unpublished; circulated by the Language Office of South Sulawesi (Balai Bahasa Sulawesi Selatan).
- Friberg, Timothy, & Friberg, Barbara. 1991. Notes on Konjo phonology. *Studies in Sulawesi linguistics, part II*, 71–115.
- Gafos, Adamantios, & Lombardi, Linda. 1999. Consonant transparency and vowel echo. *Page 8 of: North East Linguistics Society*, vol. 29.
- Ghini, Mirco. 1993. Phi-formation in Italian: A new proposal. *Toronto working papers in linguistics*, 12(2).
- Göbbel, Edward. 2007. Extraposition as PF movement. *Proceedings of WECOL 2006*, 132–145.
- Gouskova, Maria. 2012. Unexceptional segments. *Natural Language & Linguistic Theory*, 30(1), 79–133.

- Gouskova, Maria, & Bobaljik, Jonathan D. 2020. Allomorphy and vocabulary insertion. *Submitted to Handbook of Distributed Morphology, CUP*.
- Grimes, Charles E, & Grimes, Barbara D. 1987. *Languages of south Sulawesi*. Dept. of Linguistics, Research School of Pacific Studies, The Australian National University.
- Hale, Kenneth, & Selkirk, Elisabeth. 1987. Government and tonal phrasing in Papago. *Phonology*, 4, 151–183.
- Hall, T Alan. 1999. Phonotactics and the prosodic structure of German function words. *Amsterdam Studies in the Theory and History of Linguistic Science, Series IV*, 99–132.
- Halle, Morris, & Marantz, Alec. 1993. Distributed morphology and the pieces of inflection. *Pages 111–176 of: Hale, K., & Keyser, S. J. (eds), The View from Building 20*. Cambridge, MA: MIT Press.
- Halpern, Aaron. 1995. *On the placement and morphology of clitics*. Center for the Study of Language (CSLI).
- Halpern, Aaron Lars. 1992. *Topics in the placement and morphology of clitics*. Ph.D. thesis, Stanford University.
- Hamlaoui, Fatima, & Szendrői, Kriszta. 2015. A flexible approach to the mapping of intonational phrases. *Phonology*, 32(1), 79–110.
- Hamlaoui, Fatima, & Szendrői, Kriszta. 2017. The syntax-phonology mapping of intonational phrases in complex sentences: A flexible approach. *Glossa: a journal of general linguistics*, 2(1).
- Hargus, Sharon, & Tuttle, Siri G. 1997. Augmentation as affixation in Athabaskan languages. *Phonology*, 14(2), 177–220.
- Hayes, Bruce. 1981. *A metrical theory of stress rules*. Ph.D. thesis, MIT.
- Hayes, Bruce. 1989. The prosodic hierarchy in meter. *Pages 201–260 of: Rhythm and meter*. Elsevier.
- Hayes, Bruce, & Lahiri, Aditi. 1991. Bengali intonational phonology. *Natural language & linguistic theory*, 9(1), 47–96.
- Henderson, Robert. 2012. Morphological alternations at the intonational phrase edge. *Natural Language & Linguistic Theory*, 30(3), 741–787.
- Hewett, Matthew. 2023. Allomorphy in Semitic discontinuous agreement: Evidence for a modular approach to postsyntax. *Natural Language & Linguistic Theory*, 41(3), 1091–1145.
- Inkelas, Sharon, & Zec, Draga. 1990. *The phonology-syntax connection*. University of Chicago Press.
- Inkelas, Sharon, & Zec, Draga. 1995. The phonology-syntax interface. *In: Goldsmith, John A (ed), Handbook of Phonological Theory*. Blackwell Publishing.
- Ishihara, Shinichiro, & Kalivoda, Nick. 2022. Match Theory: An overview. *Language and Linguistics Compass*, 16(1), 1–26.
- Itô, Junko, & Mester, Armin. 2007. Prosodic adjunction in Japanese compounds. *MIT working papers in linguistics*, 55, 97–111.
- Itô, Junko, & Mester, Armin. 2009. The extended prosodic word. *Phonological domains: Universals and deviations*, 135–194.

- Itô, Junko, & Mester, Armin. 2012. Recursive prosodic phrasing in Japanese. *Prosody matters: Essays in honor of Elisabeth Selkirk*, 280–303.
- Itô, Junko, & Mester, Armin. 2013. Prosodic subcategories in Japanese. *Lingua*, **124**, 20–40.
- Itô, Junko, & Mester, Armin. 2019. Match as Syntax-Prosody MAX/DEP: Prosodic Enclisis in English. *English linguistics*, **36**(1).
- Ito, Junko, & Mester, Armin. 2021. Recursive prosody and the prosodic form of compounds. *Languages*, **6**(2), 65.
- Jukes, Anthony. 2006. *Makassarese (basa Mangkasara') A description of an Austronesian language of South Sulawesi*. Ph.D. thesis, The University of Melbourne.
- Kager, René Willibrord Joseph. 1996. On affix allomorphy and syllable counting. *Studia grammatica*, **41**, 155–171.
- Kaufman, Daniel. 2008. South Sulawesi pronominal clitics: Form, function and position. *Studies in Philippine languages and cultures*, **17**, 13–65.
- Kawahara, Shigeto. 2007. Copying and spreading in phonological theory: Evidence from echo epenthesis. *UMOP*, **32**, 111–144.
- King, Harold V. 1970. On blocking the rules for contraction in English. *Linguistic Inquiry*, **1**(1), 134–136.
- King, Tracy Holloway. 1996. Slavic clitics, long head movement, and prosodic inversion. *Journal of Slavic linguistics*, 274–311.
- Klein, Thomas B. 2003. Syllable structure and lexical markedness in creole morphophonology: Determiner allomorphy in Haitian and elsewhere. *The phonology and morphology of Creole languages*, 209–228.
- Kramer, Ruth. 2010. The Amharic definite marker and the syntax–morphology interface. *Syntax*, **13**(3), 196–240.
- Kubozono, Haruo. 1989. Syntactic and rhythmic effects on downstep in Japanese. *Phonology*, **6**(1), 39–67.
- Kubozono, Haruo. 1993. *The Organization of Japanese Prosody*. Kurosio Publishers.
- Kurusu, Kazutaka. 2001. *The phonology of morpheme realization*. Ph.D. thesis, University of California, Santa Cruz.
- Kusmer, Leland. 2019. *Optimal linearization: Prosodic displacement in Khoekhoegowab and beyond*. Ph.D. thesis, University of Massachusetts, Amherst.
- Kusmer, Leland Paul. 2020. Optimal Linearization: Word-Order Typology with Violable Constraints. *Syntax*, **23**(4), 313–346.
- Ladd, D Robert. 1986. Intonational phrasing: the case for recursive prosodic structure. *Phonology*, **3**, 311–340.
- Laskowske, Douglas. 2016. *Voice in Bugis: an RRG perspective*. MA Thesis, University of North Dakota.
- Lee, Jason Kwok Loong. 2008. Transitivity, Valence and Voice in Mandar. *Studies in Philippine Languages and Cultures*, **19**, 55–66.

- Liberman, Mark, & Prince, Alan. 1977. On stress and linguistic rhythm. *Linguistic inquiry*, **8**(2), 249–336.
- López, Luis. 2009a. *A derivational syntax for information structure*. Vol. 23. Oxford University Press, USA.
- López, Luis. 2009b. Ranking the linear correspondence axiom. *Linguistic Inquiry*, **40**(2), 239–276.
- Mascaró, Joan. 1996a. External allomorphy and contractions in Romance. *Probus*, **8**, 181–205.
- Mascaró, Joan. 1996b. External allomorphy as emergence of the unmarked. *Page 473–483 of: Durand, Jacques, & Laks, Bernard (eds), Current trends in phonology: Models and methods*. Salford, Manchester: University of Salford, European Studies Research Institute. Reprinted in John J. McCarthy (ed.). 2004. *Optimality Theory in Phonology: A reader*, 513–522. Oxford: Blackwell.
- Mascaró, Joan. 1996c. External allomorphy as emergence of the unmarked. *Page 473–483 of: Durand, J., & Laks, B. (eds), Current trends in Phonology: Models and Methods, vol. 2*. Salford: CNRS-ESRI.
- Mascaró, Joan. 2007. External allomorphy and lexical representation. *Linguistic Inquiry*, **38**(4), 715–735.
- Matti, David F. 1994. Mamasa pronoun sets. *Studies in Sulawesi Linguistics*, **3**, 65–89.
- McCarthy, John J. 1998. Morpheme structure constraints and paradigm occultation. *Pages 123–150 of: Gruber, M. Catherine, Higgins, Derrick, Olson, Kenneth, & Wosocki, Tamra (eds), Proceedings of CLS 32 Part 2: The Panels*. Chicago, IL: Chicago Linguistic Society.
- McCarthy, John J. 2011. Pausal phonology and morpheme realization. *Pages 341–373 of: Borowsky, Toni, Kawahara, Shigeto, Shinya, Takahito, & Sugahara, Mariko (eds), Prosody matters: Essays in honor of Lisa Selkirk*. Sheffield: Equinox.
- McCarthy, John J., & Prince, Alan. 1993. Generalized alignment. *Pages 79–153 of: Yearbook of morphology 1993*. Springer.
- Mester, R Armin. 1994. The quantitative trochee in Latin. *Natural Language & linguistic theory*, **12**(1), 1–61.
- Mills, Roger F. 1975. *Proto South Sulawesi and Proto Austronesian Phonology*. Ph.D. thesis, University of Michigan.
- Mithun, Marianne, & Basri, Hasan. 1986. The phonology of Selayarese. *Oceanic Linguistics*, **25**(1/2), 210–254.
- Moro, Andrea. 2000. *Dynamic antisymmetry*. Cambridge, MA: MIT Press.
- Muthalib, Abdul, & Sangi, M Zain. 1991. *Puisi Kalindaqdaq Mandar*. Departemen Pendidikan dan Kebudayaan.
- Myler, Neil. 2013. Linearization and post-syntactic operations in the Quechua DP. *Pages 171–210 of: Challenges to linearization*. Mouton de Gruyter Berlin.
- Myrberg, Sara. 2010. *The intonational phonology of Stockholm Swedish*. Ph.D. thesis, Acta Universitatis Stockholmiensis.
- Myrberg, Sara. 2013. Sisterhood in prosodic branching. *Phonology*, **30**(1), 73–124.
- Nespor, Marina, & Vogel, Irene. 1986. *Prosodic phonology*. De Gruyter Mouton.
- Noyer, Rolf. 1994. Mobile affixes in Huave: Optimality and morphological wellformedness. *Pages 67–82 of: Proceedings of the Twelfth West Coast Conference on Formal Linguistics*. Stanford: CSLI.

- Ostrove, Jason. 2018. Stretching, spanning, and linear adjacency in Vocabulary Insertion. *Natural Language & Linguistic Theory*, **36**(4), 1263–1289.
- Pater, Joe. 1999. Austronesian nasal substitution and other *NC effects. *Pages 310–343 of: Kager, René, van der Hulst, Harry, & Zonneveld, Wim (eds), The prosody-morphology interface*. Cambridge University Press.
- Pater, Joe. 2000. Non-uniformity in English secondary stress: the role of ranked and lexically specific constraints. *Phonology*, **17**(2), 237–274.
- Pater, Joe. 2006. The locus of exceptionality: Morpheme-specific phonology as constraint indexation. *Page 259–296 of: Bateman, Leah, O’Keefe, Michael, Reilly, Ehren, & Werle, Adam (eds), Papers in Optimality Theory III*. Amherst, MA: Graduate Linguistics Student Association, University of Massachusetts.
- Pater, Joe. 2008. Morpheme-specific phonology: Constraint indexation and inconsistency resolution. *In: Parker, Steve (ed), Phonological argumentation: Essays on evidence and motivation*. London: Equinox.
- Pelenkahu, RA, Muthalib, Abdul, & Sangi, M Zain. 1983. *Struktur Bahasa Mandar*. Pusat Pembinaan dan Pengembangan Bahasa, Departemen Pendidikan dan Kebudayaan.
- Peperkamp, Sharon Andrea. 1997. *Prosodic words*. Vol. 34. Holland Academic Graphics The Hague.
- Perlmutter, David M. 1998. Interfaces: explanation of allomorphy and the architecture of grammars. *Pages 307–338 of: Lapointe, Stephen, Brentari, Diane, & Farrel, Patrick (eds), Morphology and its relation to phonology and syntax*. Palo Alto, CA: Stanford Center for the Study of Language and Information Publications.
- Potsdam, Eric. 2022. Malagasy extraposition: Evidence for PF movement. *Natural Language & Linguistic Theory*, **40**(1), 195–237.
- Prieto, Pilar. 2005. Syntactic and eurhythmic constraints on phrasing decisions in Catalan. *Studia Linguistica*, **59**(2-3), 194–222.
- Prince, Alan, & Smolensky, Paul. 2004. *Optimality Theory: Constraint interaction in generative grammar*. John Wiley & Sons.
- Raffelsiefen, Renate. 1996. Gaps in word formation. *Pages 194–209 of: Interfaces in phonology*. Akademie Verlag.
- Raffelsiefen, Renate. 1999. *Phonological constraints on English word formation*. Springer.
- Raffelsiefen, Renate. 2004. Absolute ill-formedness and other morphophonological effects. *Phonology*, **21**(1), 91–142.
- Roehrs, Dorian. 2010. Demonstrative-reinforcer constructions. *The Journal of Comparative Germanic Linguistics*, **13**, 225–268.
- Ross, John Robert. 1967. *Constraints on variables in syntax*. Ph.D. thesis, MIT.
- Rubach, Jerzy, & Booij, Geert E. 2001. Allomorphy in optimality theory: Polish iotation. *Language*, 26–60.
- Schütze, Carson. 1994a. Serbo-Croatian second position clitic placement and the phonology-syntax interface. *MIT working papers in linguistics*, **21**, 373–473.

- Schütze, Carson. 1994b. Serbo-Croatian second position clitic placement and the phonology-syntax interface. *Pages 373–473 of: Toman, Jindrich (ed), Annual Workshop on Formal Approaches to Slavic Linguistics: The College Park Meeting*. Michigan Slavic Materials: Ann Arbor.
- Selkirk, Elisabeth. 1984. *Phonology and syntax: the relation between sound and structure*. Ph.D. thesis, MIT.
- Selkirk, Elisabeth. 1986. On derived domains in sentence phonology. *Phonology*, **3**, 371–405.
- Selkirk, Elisabeth. 1995a. The prosodic structure of function words. *University of Massachusetts occasional papers in linguistics*, **18**, Urbanczyk.
- Selkirk, Elisabeth. 1995b. Sentence prosody: Intonation, stress, and phrasing. *The handbook of phonological theory*, **1**, 550–569.
- Selkirk, Elisabeth. 2009. On clause and intonational phrase in Japanese: The syntactic grounding of prosodic constituent structure. *Gengo Kenkyu*, **136**, 35–73.
- Selkirk, Elisabeth. 2011. The syntax-phonology interface. *The handbook of phonological theory*, **2**, 435–483.
- Selkirk, Elisabeth, & Elordieta, Gorka. 2010. The role for prosodic markedness constraints in phonological phrase formation in two pitch accent languages. *Tone and Intonation in Europe*, **4**, 9–11.
- Shih, Shu-hao. 2016. Binariness and focus in prosodic phrasing: New evidence from Taiwan Mandarin. *In: Proceedings of the Annual Meetings on Phonology*.
- Sikki, Muhammad, Muthalib, Abdul, Mulya, Abdul Kadir, & Haddade, Muhammad Naim. 1987. *Kata Tugas Bahasa Mandar*. Pusat Pembinaan dan Pengembangan Bahasa, Departemen Pendidikan dan Kebudayaan.
- Snedeker, Jesse, & Trueswell, John. 2003. Using prosody to avoid ambiguity: Effects of speaker awareness and referential context. *Journal of Memory and language*, **48**(1), 103–130.
- Stanton, Juliet, & Zukoff, Sam. 2018. Prosodic identity in copy epenthesis. *Natural Language & Linguistic Theory*, **36**(2), 637–684.
- Stroik, Thomas. 2009. *Locality in minimalist syntax*. Cambridge, MA: MIT Press.
- Strømme, Kari K. 1994. Person marking in the Mamuju language. *Studies in Sulawesi Linguistics*, **3**, 91–113.
- Tranel, Bernard. 1996. French liaison and elision revisited: A unified account within Optimality Theory. *Pages 433–455 of: Parodi, Claudia, Quicoli, Carlos, Saltarelli, Mario, & Zubizarreta, Maria Luisa (eds), Aspects of Romance Linguistics: Selected Papers from the Linguistic Symposium on Romance Languages XXIV, March 10–13, 1994*. Washington, DC: Georgetown University Press.
- Truckenbrodt, Hubert. 1995. *Phonological phrases—their relation to syntax, focus, and prominence*. Ph.D. thesis, Massachusetts Institute of Technology.
- Truckenbrodt, Hubert. 1999. On the relation between syntactic phrases and phonological phrases. *Linguistic inquiry*, **30**(2), 219–255.
- Truckenbrodt, Hubert. 2002. Variation in p-phrasing in Bengali. *Linguistic Variation Yearbook*, **2**(1), 259–303.
- Truckenbrodt, Hubert. 2007. The syntax-phonology interface. *The Cambridge handbook of phonology*, **435–456**.

- Tyler, Matthew. 2019. Simplifying MATCH WORD: Evidence from English functional categories. *Glossa: a journal of general linguistics*, 4(1).
- Valkama, Kari. 1995a. Person marking in Duri. *Studies in Sulawesi linguistics*, 4, 47–95.
- Valkama, Susanne. 1995b. Notes on Duri transitivity. *Studies in Sulawesi Linguistics part IV: NUSA*, 37, 2–45.
- Van Handel, Nicholas. 2021. Matching overtly headed syntactic phrases in Italian. *Phonology*, 38(2), 317–356.
- Wagner, Michael. 2005. *Prosody and recursion*. Ph.D. thesis, Massachusetts Institute of Technology.
- Wagner, Michael. 2010. Prosody and recursion in coordinate structures and beyond. *Natural Language & Linguistic Theory*, 28(1), 183–237.
- Watson, Duane G, Gunlogson, Christine A, & Tanenhaus, Michael K. 2006. Online methods for the investigation of prosody. *Methods in empirical prosody research*, 259–282.
- Werle, Adam. 2009. *Word, phrase, and clitic prosody in Bosnian, Serbian, and Croatian*. Ph.D. thesis, University of Massachusetts Amherst.
- Yu, Alan. 2007. *A natural history of infixation*. Vol. 15. OUP Oxford.
- Zec, Draga. 2005. Prosodic differences among function words. *Phonology*, 22(1), 77–112.

Notes

¹GLOSSING: ABS: absolutive, ANTIP: antipassive, ERG: ergative, GEN: genitive, PASS: passive.

²This requirement persists beyond the system of deixis. Mandarin requires the demonstrative *do* to surface above many types of nominals that are discourse-anaphoric, and in this context, it continues to require *o*, even though it is semantically anomalous to introduce regular locative adverbs into DPs of this type (85a)–(85b). The regular locative adverbs are never required in any corresponding way: they are always optional and they can always co-occur with minimally mismatched demonstratives, such that *indi* and *dini* can surface beneath *ndi* and *de* and *dio* and *diting* can surface beneath *do* and *iting* (85c)–(85d).

(85) *Selection Beyond Deixis*

- a. Diang da'dua analisa: mesa sittassis, mesa ponologi
 there are two analysis one syntactic one phonological
 'There are two analyses: one syntactic, one phonological.'
- b. Macoa i [_{DP} do analisa ponologi (#dio) *(o)].
 good 3ABS that analysis phonological there there
 'The phonological analysis (#there) is elegant.'

- | | |
|--|---|
| c. De posa [indi / dini / ____] e
this cat here here here
‘This cat here’ | d. Do posa [dio / diting / ____] o
that cat there there there
‘That cat there’ |
|--|---|

³In the investigation that follows, we will restrict our attention to the type of prosody that emerges most readily when speakers are asked to introspect on the most natural pronunciation of particular strings. Following the methodology of Nespor & Vogel 1986 and my prior work on Mandar (Brodkin, 2025a), I will thus abstract away from the many factors that can derail patterns of prosodic phrasing in the context of natural speech (Snedeker & Trueswell, 2003; Aylett & Turk, 2004; Watson *et al.*, 2006; Féry & Ishihara, 2016).

⁴Many other systems in the phonology of the language converge with these two effects to single out the same domain: for instance, the edges that block Nasal Assimilation and host Comma Intonation also (i) host reinforcers (this section), (ii) license the emergence of special types of feet (Section 5); (iii) license the emergence of particular glottal segments; (iv) host the types of intonational contours canonically associated with the *ι*, and (v) host a series of “final particles” that descriptively associate with contours of this type.

⁵For discussion of several other cases of putative phonological displacement, which targets elements and landing sites that seem more easily defined in syntactic terms, see Göbbel 2007; López 2009a; Werle 2009; Agbayani & Golston 2010; Agbayani *et al.* 2015; Agbayani & Golston 2016; Clemens & Coon 2018; Clemens 2019; Potsdam 2022.

⁶This description breaks from the traditional analysis of word-level stress in the language, which takes it to fall on the penult without regard to syllable weight. This quantity-insensitive analysis has been asserted and advanced in the grammar of Mandar by R.A Pelenkahu, Abdul Muthalib, and Zain Sangi (Pelenkahu *et al.*, 1983), in several of my own works (Brodkin 2025b) and in every discussion of stress that I am aware of in every other language of South Sulawesi (Friberg & Friberg 1991; Jukes 2006, and especially Mithun & Basri 1986; Basri *et al.* 1999; Basri 1999; Basri *et al.* 2012, a series of works on Selayarese that are co-authored by Hasan Basri, a native speaker of that language). Over my first seven years of hearing, speaking, and working on Mandar, I assumed this analysis erroneously for six reasons: (i) I was unwilling to make judgments about metrical structure on the basis of my impressions about phonetic prominence (though the rightmost trochee in the word clearly hosts word-level stress (Pelenkahu *et al.*, 1983) and I believe that an audible secondary stress falls on the head of each trochee to the left); (ii) word-final monosyllabic feet are ruled out in many positions by the Rhythm Rule (Footnote X) and the Positional Branching Constraint (this subsection), (iii) word-final monosyllabic feet are favored at the right edge of the intonational phrase (next subsection), (iv) the right edge of the intonational phrase hosts so many independent phonetic distractions and higher-level

phonological events that it is difficult to single out word-level prominence in that position, (v) the system of footing is derailed in complex ways by the phonology of focus, which interacts with every other system in the phrasal phonology as well, and (vi) the interactions between phonetic prominence, word-level metrical structure, and top-down constraints are so complex in Indonesian, the language of contact for this work, that I have been slow to interpret my own intuitions about Mandar.

⁷We can see the same effect across all of the classes of monosyllabic heads studied in [Brodkin 2025b](#). The native adverbs *bo* ‘again’ and *to* ‘also’ to begin, are monosyllabic in phrase-medial positions, disyllabic at the right edges of medial phonological phrases that are final in the maximal phonological phrase (86a), and monosyllabic again at the right edge of the intonational phrase (86b).

(86) *Native Monosyllables: Epenthesis Suspended*

- | | |
|--|---|
| <p>a. bukúna^H wémme (wóʔo)^H maníni^H.
 buku-na bemmeʔ bo manini
 book-3GEN fall again later
 ‘The book will fall again later.’</p> | <p>b. bukúna^H wémme (wóʔ)^H.
 buku-na bemmeʔ bo
 book-3GEN fall again
 ‘The book will fall again.’</p> |
|--|---|

The same effects can be seen with the preposition *sung* ‘out’ and the negator *da* ‘don’t.’ Like all functional heads in the language, these elements are typically parsed as stray syllables and directly integrated into the higher prosodic domains built around their complements (and see also [Selkirk 1995a](#); [Peperkamp 1997](#); [Booij 1999](#); [Hall 1999](#); [Itô & Mester 2009](#); [Tyler 2019](#)). When stranded by complement ellipsis, they are forced to fall at the edges of phonological phrases that correspond to the projections they head (and see [Anderson 2008](#); [Itô & Mester 2019](#); [Brodkin 2025b](#), as well as [King 1970](#)). In this position, they undergo ʔv-epenthesis at the right edges of medial phonological phrases that are final in the maximal phonological phrase but cease to undergo this repair at the edge of the intonational phrase.

(87) *Monosyllabic Functional Heads: Parallel Effects*

- | | |
|---|---|
| <p>a. bémme^H súʔun^H diyénaʔ.
 bemmeʔ i suŋ ____ digenaʔ
 fall 3ABS out NP earlier
 ‘It fell out earlier.’</p> | <p>b. bémme^H (súŋ)^H.
 bemmeʔ i suŋ
 fall 3ABS out
 ‘It fell out.’</p> |
|---|---|

⁸When we rule out Syllable Attraction in this second configurations, the reinforcers lose the ability to remain in-situ. In VSO strings where the object bears initial stress, the reinforcers that are generated in the s are forced to move (??) or delete (??). In the complex DPs where their initial positions are followed by specifiers that begin with stressed syllables or host two prosodic words, we see the same effects (88a)-(88d).

(88) *Phonological Disruptions*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
 máne wénjaŋ^H i do wúku^H — (yúru)^H diyénaʔ^H óʔ^H.
 mane u-beŋjaŋ i do buku guru-u digenaʔ o
 just 1ERG-give 3ABS that book teacher earlier there
 ‘I just gave that book there to my teacher earlier.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
 máne wénjaŋ^H i do wúku^H — (yúru)^H díní^H éʔ^H.
 mane u-beŋjaŋ i do buku o guru-u dini e
 just 1ERG-give 3ABS that book there teacher here here
 ‘I just gave that book there to my teacher here.’
- c. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
 do wukúna^H — (yúru)^H dó .i méja^H óʔ^H.
 do buku-na guru dio di medʒa o
 that book-3GEN teacher there on table there
 ‘That book of the teacher’s there on the table.’
- d. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
 do wukúna^H — yu(rúna)^H ikátʃoʔ díní .i méja^H éʔ^H.
 do buku-na o guru-na ikátʃoʔ dini di medʒa e
 that book-3GEN there teacher-3GEN NAME here on table here
 ‘That book of the teacher’s here on the table.’

⁹The same effect is forced in phrase-medial positions by the Rhythm Rule, which prevents monosyllabic feet from surfacing before words that bear initial stress. Examples (89a) shows its basic effect with a final-cvccvc root: before the adverb *bandi* ‘really’, it takes penultimate stress and hosts a final disyllabic foot. Example (89b) then shows how the suffix *i* reacts to this constraint: in this prosodic context, it is suppressed.

(89) *The Rhythm Rule*

- a. $\{\phi\}$ matʃóa (sánna) (wándi) }^H $\{\phi\}$ óʔ
 matʃoa sanna^l bandi o
 good very really there
 }^H ‘Is it really very good there?’
- b. $\{\phi\}$ móloʔ — (wándi) }^H $\{\phi\}$ óʔ }^H
 mu-oloʔ-i bandi o
 2ERG-like-I really there
 ‘Do you really like it there?’

¹⁰These results turn on two further hypotheses about the inventory of possible constraints. First, I assume that the interface cannot detect the boundaries of syntactic islands in the morphological representations it receives. Second, I assume that the interface cannot leverage c-command relationships in the input to restrict movement to phonologically defined positions. These hypotheses rule out two types of constraints: those that would prohibit phonological relinearization out of syntactic islands and those that would force phonological relinearization to select its targets on the basis of relative c-command.